

Convective Heat Transfer

2.1. DIMENSIONAL ANALYSIS

Dimensional analysis is a mathematical methods which makes use of the study of the dimensions for solving several engineering problems. This methods can be applied to all types of fluid resistances, heat flow problems and many other problems in fluid mechanics and thermodynamics.

2.1.1. Dimensions

In dimensional analysis, the various physical quantities used in fluid phenomenon can be expressed in terms of fundamental quantities. These fundamental quantities are mass (M), length (L), time (T), and temperature (θ)

The dimensions of commonly used quantities in heat transfer analysis is listed in Table 2.1 with reference to ML θ T where

M	=	Mass,
L	=	Length,
Theta	=	Temperature,
T	=	Time.

For example,

$$\text{Velocity } V = \frac{\text{Distance}}{\text{Time}} = \frac{L}{T} = LT^{-1}$$

Table 2.1.

S. No.	Quantity	Symbol	Units (SI)	Dimensions (M, L, T, θ system)
1.	Length	L	m	L
2.	Area	A	m ²	L ²
3.	Velocity	v	m/s	LT ⁻¹
4.	Acceleration	a, g	m/s ²	LT ⁻²
5.	Mass	m	kg	M
6.	Density	ρ	kg/m ³	ML ⁻³
7.	Force	F	N	MLT ⁻²
8.	Pressure	p	N/m ²	ML ⁻¹ T ⁻²
9.	Work	W	J	ML ² T ⁻²
10.	Torque	T	J	ML ² T ⁻²
11.	Power	P	W	ML ² T ⁻³
12.	Kinematic viscosity	ν	m ² /s	L ² T ⁻¹
13.	Discharge	Q	m ³ /s	L ³ T ⁻¹
14.	Heat	Q	J	ML ² T ⁻²
15.	Heat transfer coefficient	h	W/m ² K	MT ⁻³ θ^{-1}
16.	Thermal conductivity	k	W/mK	MLT ⁻³ θ^{-1}
17.	Thermal diffusivity	α	m ² /s	L ² T ⁻¹
18.	Specific heat	C _p	J/kg K	L ² T ⁻² θ^{-1}
19.	Heat transfer rate	q	W	ML ² T ⁻³
20.	Coefficient of expansion	β	K ⁻¹	θ^{-1}

2.1.2. Buckingham π Theorem

A more general situation in which dimensional analysis may be profitably employed is one in which there is no governing differential equation clearly applies. In such a situation, a more general procedure is required which is known as Buckingham π theorem.

Buckingham π theorem states as follows.

“If there are n variables in a dimensionally homogeneous equation and if these contain m fundamental dimensions, then the variables are arranged into $(n - m)$ dimensionless terms. These dimensionless term are called π terms.”

2.1.3. Advantages of Dimensional Analysis

1. It expresses the functional relationship between the variables in dimensionless terms.
2. It enables getting up a theoretical solution in a simplified dimensionless form.
3. Design curves, by the use of dimensional analysis, can be developed from the experimental data or direct solution of the problem.
4. The result of one series of tests can be applied to a large number of other similar problems with the help of dimensional analysis.

2.1.4. Limitations of Dimensional Analysis

1. The complete information is not provided by dimensional analysis. It only indicates that there is some relationship between the parameters.
2. No information is given about the internal mechanism of the physical phenomenon.
3. Dimensional analysis does not give any clue regarding the selection of variables.

2.2. DIMENSIONLESS NUMBERS AND THEIR PHYSICAL SIGNIFICANCE

2.2.1. Reynolds Number (Re)

It is defined as the ratio of inertia force to viscous force.

$$\begin{aligned} \text{Re} &= \frac{\text{Inertia Force}}{\text{Viscous Force}} \\ &= \frac{\rho U^2 L^2}{\mu UL} \\ \text{Re} &= \frac{UL}{\nu} \end{aligned} \quad \dots(2.1)$$

Where U - Velocity, m/s,
 L - Length, m,

$$V = \frac{\mu}{\rho} \quad - \quad \text{Kinematic Viscosity, m}^2/\text{s}.$$

Reynolds number, is therefore, a measure of relative magnitude of the inertia force to the viscous force occurring in the flow.

2.2.2. Prandtl Number (Pr)

It is the ratio of the momentum diffusivity to the thermal diffusivity.

$$\begin{aligned} \text{Pr} &= \frac{\text{Momentum diffusivity}}{\text{Thermal diffusivity}} \\ \text{Pr} &= \frac{\mu C_p}{k} = \frac{V}{\alpha} \quad \dots(2.2) \end{aligned}$$

Where

$$\begin{aligned} V &= \text{Kinematic Viscosity, m}^2/\text{s}, \\ \alpha &= \text{Thermal diffusivity, m}^2/\text{s}. \end{aligned}$$

Prandtl number provides a measure of the relative effectiveness of the momentum and energy transport by diffusion.

2.2.3. Nusselt Number (Nu)

It is defined as the ratio of the heat flow by convection process under an unit temperature gradient to the heat flow rate by conduction under an unit temperature gradient through a stationary thickness of L metre.

$$\begin{aligned} \text{Nusselt Number (Nu)} &= \frac{q_{conv}}{q_{conv}} = \frac{h A \Delta T}{k A \frac{\Delta T}{L}} = \frac{h}{\frac{k}{L}} \\ (\text{Nu}) &= \frac{hL}{k} \quad \dots(2.3) \end{aligned}$$

Where

$$\begin{aligned} h &- \text{Heat transfer coefficient W/m}^2\text{k}, \\ L &- \text{Length, m}, \\ K &- \text{Thermal conductivity, W/mK}. \end{aligned}$$

The Nusselt number is a convenient measure of the convective heat transfer coefficient. For a given value of the Nusselt Number, the convective heat transfer coefficient is directly proportional to thermal conductivity of the fluid and inversely proportional to the significant length.

2.2.4. Grashof Number (Gr)

It is defined as the ratio of product of inertia force and buoyancy force to the square of viscous force.

$$\begin{aligned}
 Gr &= \frac{\text{Inertia force} \times \text{Buoyancy force}}{(\text{Viscous force})^2} \\
 Gr &= \frac{\rho U^2 L^2 \times \rho \beta g \Delta T L^3}{(\mu U L)^2} \\
 &= \frac{g \rho^2 \beta L^3 \Delta T}{\mu^2} \\
 Gr &= \frac{g \times \beta \times L^3 \times \Delta T}{\nu^2} \quad \dots(2.4)
 \end{aligned}$$

Where β - Coefficient of expansion, K^{-1}
 L - Length, m,
 ν - kinematic viscosity, m^2/s ,
 ΔT - Temperature difference, K.

Grashof Number has a role in free convection similar to that played by Reynolds number in forced convection.

2.2.5. Stanton Number (St)

It is the ratio of Nusselt Number to the product of Reynolds number and Prandtl Number.

$$\begin{aligned}
 St &= \frac{Nu}{Re \times Pr} = \frac{\frac{h L}{k}}{\frac{U L}{\nu} \times \frac{\mu C_p}{k}} \\
 &= \frac{\frac{h L}{k}}{\frac{\rho U L}{\mu} \times \frac{\mu C_p}{k}} \\
 St &= \frac{h}{\rho U C_p}
 \end{aligned}$$

2.2.6. Newtonian and Non-Newtonian fluids

The fluids which obey the Newton's law of viscosity are called the Newtonian fluids and those which do not obey are called non-newtonian fluids.

2.2.7. Laminar Flow

Laminar flow is sometimes also called stream line flow. In this type of flow, the fluid moves in layers and each fluid particles follows a smooth and continuous path. The fluid particles in each layer remain in an orderly sequence without making with each other.

2.2.8. Turbulent Flow

In addition to the laminar type of flow, a distinct irregular flow is frequently observed in nature. This type of flow is called turbulent flow. The path of any individual particles is Zig-Zag and irregular. Fig 2.1 shows the instantaneous velocity in laminar and turbulent flow.

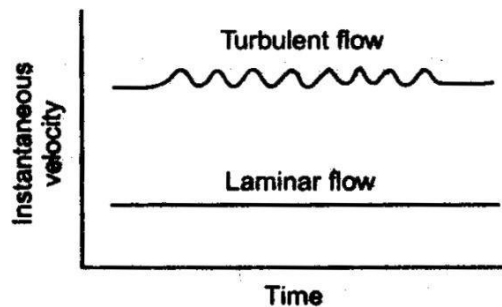


Fig. 2.1.

2.3. Boundary layer concept

The concept of a boundary layer as proposed by Prandtl forms the starting point for the simplification of the equation of motion and energy.

When a real fluid *i.e.*, Viscous fluid, flows along a stationary solid boundary, a layer of fluid which comes in contact with the boundary surfaces. Thus the layer of fluid which cannot slip away the boundary surfaces and undergoes retardation. This retarded layer further causes retardation for the adjacent layer of the fluid. So, small region is developed in the immediate vicinity of the boundary surfaces in which the velocity of the flowing fluid increases rapidly from zero at boundary surfaces and approaches the velocity of main stream.

The layer adjacent to the boundary is known as boundary layer. Boundary layer is formed whenever there is relative motion between the boundary and the fluid.

In this concept, the flow over a body is divided into two regions.

1. A thin region near the body called the boundary layer, where the velocity and temperature gradients are large.
2. The region outside the boundary layer where velocity and temperature gradients are very nearly equal to their free stream values.

The thickness of the boundary layer has been defined as the distance from the surface at which the local velocity or temperature reaches 99% of the external velocity or temperature.

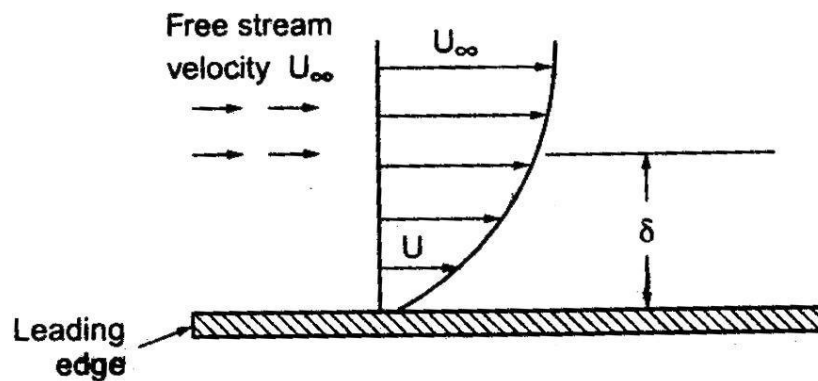


Fig. 2.2. Boundary layer on flat plate

2.3.1. Types of Boundary Layer

1. Hydrodynamic boundary layer (or) Velocity boundary layer
2. Thermal boundary layer

2.3.2. Hydrodynamic Boundary Layer

In hydrodynamic boundary layer, velocity of the fluid is less than 99% of free stream velocity.

2.3.3. Thermal Boundary Layer

In thermal boundary layer, temperature of the fluid is less than 99% of free stream temperature.

2.4. CONVECTION

Convection is a process of heat transfer that will occur between a solid surface and a fluid medium when they are at different temperatures.

2.4.1. Newton's Law of Convection

Heat transfer from the moving fluid to solid surfaces is given by the equation,

$$Q = h A (T_w - T_\infty)$$

This equation is referred to as Newton's Law of cooling.

Where,

h	=	Local heat transfer coefficient in W/m ² K,
A	=	Surface area in m ² ,
T_w	=	Surface (or) Wall temperature in K,
T_∞	=	Temperature of the fluid in K.

2.4.2. Types of Convection

1. Free Convection,
2. Forced convection.

2.4.3. Free (or) Natural Convection

If the fluid motion is produced due to change in density resulting from temperature gradients, the mode of heat transfer is said to be free or natural convection.

2.4.4. Forced Convection

If the fluid motion is artificially created by means of an external force like a blower or fan, that type of heat transfer is known as forced convection.

2.5. THE LOCAL AND AVERAGE HEAT TRANSFER COEFFICIENTS FOR FLAT PLATE – LAMINAR FLOW

At the surface of the flat plate, heat flow may be written as

$$\begin{aligned} q &= \frac{Q}{A} = h_x (T_w - T_\infty) \\ &= k \left(\frac{\partial T}{\partial y} \right)_{y=0} \end{aligned} \quad \dots(2.6)$$

We know that,

$$\begin{aligned}
 \left(\frac{\partial T}{\partial y} \right)_{y=0} &= T_w - T_\infty \times \sqrt{\frac{U}{\nu x}} \times \left(\frac{\partial \theta}{\partial \eta} \right)_{\eta=0} \\
 &= T_w - T_\infty \times \sqrt{\frac{U}{\nu x}} \times 0.332 (\text{Pr})^{0.333} \\
 &\quad \left[\because \left(\frac{\partial \theta}{\partial \eta} \right)_{\eta=0} = 0.332 (\text{Pr})^{0.333} \right] \\
 &= T_w - T_\infty \times \sqrt{\frac{U}{\nu}} \times x^{-1/2} \times 0.332 \times \text{Pr}^{0.333} \\
 &= (T_w - T_\infty) \times \sqrt{\frac{U}{\nu}} \times x^{1/2} \times x^{-1} \times 0.332 (\text{Pr})^{0.333} \\
 &= T_w - T_\infty \times \sqrt{\frac{Ux}{\nu}} \times \frac{1}{x} \times 0.332 \times (\text{Pr})^{0.333} \\
 \left(\frac{\partial T}{\partial y} \right)_{y=0} &= \frac{T_w - T_\infty}{x} \times \sqrt{\frac{Ux}{\nu}} \times 0.332 (\text{Pr})^{0.333}
 \end{aligned}$$

Substituting $\left(\frac{\partial T}{\partial y} \right)_{y=0}$ in equation (2.6),

$$\begin{aligned}
 (2.6) \Rightarrow \quad \frac{Q}{A} &= h_x (T_w - T_\infty) = k \left(\frac{\partial T}{\partial y} \right)_{y=0} \\
 \Rightarrow \quad \frac{Q}{A} &= k \times \frac{(T_w - T_\infty)}{x} \times \sqrt{\frac{Ux}{\nu}} \times 0.332 \times (\text{Pr})^{0.333} \\
 \frac{Q}{A} &= 0.332 \times \frac{k}{x} \times (T_w - T_\infty) (\text{Re})^{1/2} \times (\text{Pr})^{0.333} \\
 &\quad \left[\because \text{Re} = \frac{Ux}{\nu} \right]
 \end{aligned}$$

$$\Rightarrow h_x (T_w - T_\infty) = 0.332 \times \frac{k}{x} (T_w - T_\infty) \times (\text{Re})^{0.5} (\text{Pr})^{0.333}$$

$$\Rightarrow h_x = 0.332 \times \frac{k}{x} \times (\text{Re})^{0.5} (\text{Pr})^{0.333}$$

We know that,

$$\boxed{\text{Local heat transfer coefficient, } h_x} = 0.332 \times \frac{k}{x} (\text{Re})^{0.5} (\text{Pr})^{0.333} \quad \dots (2.7)$$

We know that,

$$\left. \begin{array}{l} \text{Local Nusselt} \\ \text{number, } \text{Nu}_x \end{array} \right\} = \frac{h_x x}{k}$$

$$\Rightarrow \text{Nu}_x = \frac{0.332 \times \frac{k}{x} (\text{Re})^{0.5} (\text{Pr})^{0.333} \times x}{k}$$

$$\Rightarrow \boxed{\text{Local Nusselt Number } \text{Nu}_x} = 0.332 (\text{Re})^{0.5} (\text{Pr})^{0.333} \quad \dots (2.8)$$

The average heat transfer coefficient, h is given by

$$\begin{aligned} h &= \frac{1}{L} \int_0^L h_x dx \\ &= \frac{1}{L} \int_0^L 0.332 \times \frac{k}{x} (\text{Re})^{0.5} (\text{Pr})^{0.333} dx \\ &= \frac{1}{L} \int_0^L 0.332 \times \frac{k}{x} \left(\frac{Ux}{v} \right)^{0.5} (\text{Pr})^{0.333} dx \quad \left[\because \text{Re} = \frac{Ux}{v} \right] \\ &= \frac{1}{L} \times 0.332 \times k \times \left(\frac{U}{v} \right)^{0.5} \times (\text{Pr})^{0.333} \times \int_0^L \frac{1}{x} \times (x)^{0.5} dx \\ &= \frac{1}{L} \times 0.332 \times k \times \left(\frac{U}{v} \right)^{0.5} (\text{Pr})^{0.333} \times \int_0^L x^{-1} \times x^{0.5} dx \\ &= \frac{0.332}{L} \times k \times \left(\frac{U}{v} \right)^{0.5} \times (\text{Pr})^{0.333} \times \int_0^L x^{-0.5} dx \\ &= \frac{0.332}{L} \times k \times \left(\frac{U}{v} \right)^{0.5} \times (\text{Pr})^{0.333} \times \left[\frac{x^{-0.5+1}}{-0.5+1} \right]_0^L \\ &= \frac{0.332}{L} \times k \times \left(\frac{U}{v} \right)^{0.5} \times (\text{Pr})^{0.333} \times \left[\frac{L^{0.5}}{0.5} \right] \\ &= \frac{0.332}{0.5} \times \left(\frac{k}{L} \right) \times \left(\frac{UL}{v} \right)^{0.5} (\text{Pr})^{0.333} \end{aligned}$$

$$= 0.664 \left(\frac{k}{L} \right) (Re)^{0.5} (Pr)^{0.333} \quad \left[\because Re = \frac{UL}{\nu} \right]$$

$$\boxed{\text{Average heat transfer coefficient, } h} = 0.664 \left(\frac{k}{L} \right) (Re)^{0.5} (Pr)^{0.333}$$

... (2.9)

We know that,

$$\text{Average Nusselt Number, } Nu = \frac{hL}{k}$$

$$\Rightarrow Nu = \frac{0.664 \left(\frac{k}{L} \right) (Re)^{0.5} (Pr)^{0.333} \times L}{k}$$

$$\Rightarrow Nu = 0.664 (Re)^{0.5} (Pr)^{0.333}$$

$$\boxed{\text{Average Nusselt number, } Nu} = 0.664 (Re)^{0.5} (Pr)^{0.333}$$

From equation (2.7) and (2.9), we know that,

$$h = 2 h_x$$

2.6. THE LOCAL AND AVERAGE HEAT TRANSFER COEFFICIENTS FOR FLAT PLATE – TURBULENT FLOW

The heat transfer coefficient for turbulent flow can be derived by using Colburn analogy,

For colburn analogy, we know that,

$$St_x (Pr)^{2/3} = \frac{C_{fx}}{2} = \frac{0.0592}{2} (Re_x)^{-0.2}$$

[From HMT data book, Page No.113 (Sixth Edition)]

$$\Rightarrow St_x (Pr)^{2/3} = \frac{0.0592}{2} (Re_x)^{-0.2}$$

$$\Rightarrow \frac{Nu_x}{Re_x Pr} \times [Pr]^{2/3} = 0.0296 (Re_x)^{-0.2}$$

$$\Rightarrow \frac{Nu_x}{Re_x} \times (Pr)^{-1/3} = 0.0296 (Re_x)^{-0.2}$$

$$\Rightarrow Nu_x (Pr)^{-1/3} = 0.0296 \times (Re_x) (Re_x)^{-0.2}$$

$$\Rightarrow Nu_x = 0.0296 (Re_x)^{0.8} (Pr)^{1/3}$$

$$\boxed{\text{Local Nusselt Number, } Nu_x \} = 0.0296 (Re)^{0.8} (Pr)^{0.333}$$

...(2.11)

We know that,

$$\begin{aligned} Nu_x &= \frac{h_x x}{k} \\ \Rightarrow \frac{h_x x}{k} &= 0.0296 (Re_x)^{0.8} (Pr)^{0.333} \\ \Rightarrow h_x &= \frac{0.0296 (Re_x)^{0.8} \times (Pr)^{0.333} \times k}{x} \\ \Rightarrow h_x &= 0.0296 \left(\frac{k}{x} \right) (Re_x)^{0.8} (Pr)^{0.333} \end{aligned}$$

$$\boxed{\text{Local heat transfer coefficient, } h_x \} = 0.0296 \left(\frac{k}{x} \right) (Re_x)^{0.8} (Pr)^{0.333}$$

...(2.12)

The average heat transfer coefficient, h is given by

$$\begin{aligned} h &= \frac{1}{L} \int_0^L h_x dx \\ &= \frac{1}{L} \int_0^L 0.0296 \left(\frac{k}{x} \right) (Re_x)^{0.8} (Pr)^{0.333} dx \\ &= \frac{1}{L} \int_0^L 0.0296 \times \left(\frac{k}{x} \right) \times \left(\frac{Ux}{v} \right)^{0.8} (Pr)^{0.333} dx \\ &\quad \left[\because Re = \frac{Ux}{v} \right] \\ &= \frac{0.0296}{L} \times k \times \left(\frac{U}{v} \right)^{0.8} (Pr)^{0.333} \int_0^L \left(\frac{1}{x} \right) \times x^{0.8} dx \\ &= 0.0296 \times \left(\frac{k}{L} \right) \times \left(\frac{U}{v} \right)^{0.8} \times (Pr)^{0.333} \int_0^L x^{-0.2} dx \\ &= 0.0296 \times \left(\frac{k}{L} \right) \times \left(\frac{U}{v} \right)^{0.8} \times (Pr)^{0.333} \left[\frac{x^{-0.2+1}}{-0.2+1} \right]_0^L \\ &= 0.0296 \times \left(\frac{k}{L} \right) \times \left(\frac{U}{v} \right)^{0.8} \times (Pr)^{0.333} \left[\frac{L^{0.8}}{0.8} \right] \end{aligned}$$

$$\begin{aligned}
&= 0.037 \left(\frac{k}{L} \right) \left(\frac{UL}{\nu} \right)^{0.8} (\text{Pr})^{0.333} \\
&= 0.037 \left(\frac{k}{L} \right) (\text{Re})^{0.8} (\text{Pr})^{0.333} \quad \left[\because \text{Re} = \frac{UL}{\nu} \right]
\end{aligned}$$

$$\boxed{\text{Average heat transfer coefficient, } h} = 0.037 \left(\frac{k}{L} \right) (\text{Re})^{0.8} (\text{Pr})^{0.333}$$

We know that,

$$\begin{aligned}
\left. \begin{array}{l} \text{Average Nusselt} \\ \text{Number, Nu} \end{array} \right\} &= \frac{hL}{k} \\
\Rightarrow \text{Nu} &= \frac{0.037 \left(\frac{k}{L} \right) (\text{Re})^{0.8} (\text{Pr})^{0.333} \times L}{k}
\end{aligned}$$

$$\left. \begin{array}{l} \text{Average Nusselt} \\ \text{Number, Nu} \end{array} \right\} = 0.037 (\text{Re})^{0.8} (\text{Pr})^{0.333}$$

From equation (2.12) and (2.13), we know that,

$$\begin{aligned}
\left. \begin{array}{l} \text{Average heat transfer} \\ \text{coefficient, } h \end{array} \right\} &= 1.25 \times h_x \\
\Rightarrow \boxed{h} &= 1.25 h_x
\end{aligned}$$

2.6.1. Heat Transfer coefficient for combination of Laminar and Turbulent Flow

Heat transfer coefficient for laminar – turbulent combined flow is given by

$$\begin{aligned}
h &= \underbrace{\frac{1}{L} \int_0^x h_x dx}_{\text{(Laminar)}} + \underbrace{\frac{1}{L} \int_x^L h_x dx}_{\text{(Turbulent)}} \\
&= \frac{1}{L} \left[\int_0^x 0.332 \left(\frac{k}{x} \right) (\text{Re})^{0.5} (\text{Pr})^{0.333} dx + \int_x^L 0.0296 \left(\frac{k}{x} \right) (\text{Re})^{0.8} (\text{Pr})^{0.333} dx \right]
\end{aligned}$$

$$\begin{aligned}
&= \frac{1}{L} \left[\int_0^x 0.332 \left(\frac{k}{x} \right) \left(\frac{Ux}{v} \right)^{0.5} (Pr)^{0.333} dx + \right. \\
&\quad \left. \int_x^L 0.0296 \left(\frac{k}{x} \right) \left(\frac{Ux}{v} \right)^{0.8} (Pr)^{0.333} dx \right] \\
&\quad \left[\because Re = \frac{Ux}{v} \right] \\
&= \frac{k}{L} (Pr)^{0.333} \left[0.332 \left(\frac{U}{v} \right)^{0.5} \int_0^x \frac{x^{0.5}}{x} dx + \right. \\
&\quad \left. 0.0296 \left(\frac{U}{v} \right)^{0.8} \int_x^L \frac{x^{0.8}}{x} dx \right] \\
&= \frac{k}{L} (Pr)^{0.333} \left[0.332 \left(\frac{U}{v} \right)^{0.5} \int_0^x x^{-0.5} dx + \right. \\
&\quad \left. 0.0296 \left(\frac{U}{v} \right)^{0.8} \int_x^L x^{-0.2} dx \right] \\
&= \frac{k}{L} (Pr)^{0.333} \left[0.332 \left(\frac{U}{v} \right)^{0.5} \left[\frac{x^{-0.5+1}}{-0.5+1} \right]_0^x + \right. \\
&\quad \left. 0.0296 \left(\frac{U}{v} \right)^{0.8} \left[\frac{x^{-0.2+1}}{-0.2+1} \right]_x^L \right] \\
&= \frac{k}{L} (Pr)^{0.333} \left[0.332 \left(\frac{U}{v} \right)^{0.5} \left[\frac{x^{0.5}}{0.5} \right] + \right. \\
&\quad \left. 0.0296 \left(\frac{U}{v} \right)^{0.8} \left[\frac{L^{0.8}}{0.8} - \frac{x^{0.8}}{0.8} \right] \right] \\
&= \frac{k}{L} (Pr)^{0.333} \left[\frac{0.332}{0.5} \left(\frac{Ux}{v} \right)^{0.5} + \frac{0.0296}{0.8} \left(\frac{UL}{v} \right)^{0.8} \right. \\
&\quad \left. - \frac{0.0296}{0.8} \left(\frac{Ux}{v} \right)^{0.8} \right]
\end{aligned}$$

$$h = \frac{k}{L} (Pr)^{0.333} \left[0.664 (Re_x)^{0.5} + 0.037 (Re_L)^{0.8} - 0.037 (Re_x)^{0.8} \right]$$

Transition occurs at critical Reynolds number, $Re_c = 5 \times 10^5$, i.e., Flow is laminar upto $Re = 5 \times 10^5$, after that flow is turbulent.

Substitute $Re_c = Re_x = 5 \times 10^5$

$$\begin{aligned} \Rightarrow h &= \frac{k}{L} (Pr)^{0.333} \left[0.664 (5 \times 10^5)^{0.5} + \right. \\ &\quad \left. 0.037 (Re_L)^{0.8} - 0.037 (5 \times 10^5)^{0.8} \right] \\ &= \frac{k}{L} (Pr)^{0.333} \left[0.037 (Re_L)^{0.8} - 871 \right] \end{aligned}$$

Average heat transfer coefficient,

$$h = \frac{k}{L} (Pr)^{0.333} \left[0.037 (Re_L)^{0.8} - 871 \right]$$

... (2.15)

We know that,

$$\text{Average Nusselt Number, } Nu = \frac{hL}{k}$$

$$\text{Average Nusselt Number, } Nu = \frac{hL}{k}$$

$$\Rightarrow Nu = \frac{\frac{k}{L} (Pr)^{0.333} \left[0.037 (Re_L)^{0.8} - 871 \right] \times L}{k}$$

$$\text{Average Nusselt Number, } Nu = Pr^{0.333} [0.037 (Re_L)^{0.8} - 871]$$

... (2.16)

2.7. BOUNDARY LAYER THICKNESS, SHEAR STRESS AND SKIN FRICTION COEFFICIENT FOR TURBULENT FLOW

We know that, Von Karman momentum equation for boundary layer flow is

$$\frac{\tau_0}{\rho U^2} = \frac{d}{dx} \left[\int_0^{\delta} \frac{u}{U} \left(1 - \frac{u}{U} \right) dy \right]$$

Substitute, $\frac{u}{U} = \left(\frac{y}{\delta} \right)^{1/7}$

$$\begin{aligned} \Rightarrow \frac{\tau_0}{\rho U^2} &= \frac{d}{dx} \left[\int_0^{\delta} \left(\frac{y}{\delta} \right)^{1/7} \left[1 - \left(\frac{y}{\delta} \right)^{1/7} \right] dy \right] \\ &= \frac{d}{dx} \left\{ \int_0^{\delta} \left[\left(\frac{y}{\delta} \right)^{1/7} - \left(\frac{y}{\delta} \right)^{2/7} \right] dy \right\} \\ &= \frac{d}{dx} \left\{ \frac{1}{\delta^{1/7}} \int_0^{\delta} y^{1/7} dy - \frac{1}{\delta^{2/7}} \int_0^{\delta} y^{2/7} dy \right\} \\ &= \frac{d}{dx} \left\{ \frac{1}{\delta^{1/7}} \left(\frac{y^{\frac{1}{7}+1}}{\frac{1}{7}+1} \right)_0^{\delta} - \frac{1}{\delta^{2/7}} \left(\frac{y^{\frac{2}{7}+1}}{\frac{2}{7}+1} \right)_0^{\delta} \right\} \\ &= \frac{d}{dx} \left\{ \frac{1}{\delta^{1/7}} \left(\frac{y^{8/7}}{8/7} \right)_0^{\delta} - \frac{1}{\delta^{2/7}} \left(\frac{y^{9/7}}{9/7} \right)_0^{\delta} \right\} \\ &= \frac{d}{dx} \left[\frac{7}{8} \left(\frac{\delta^{8/7}}{\delta^{1/7}} \right) - \frac{7}{9} \left(\frac{\delta^{9/7}}{\delta^{2/7}} \right) \right] \\ &= \frac{d}{dx} \left[\frac{7}{8} \delta - \frac{7}{9} \delta \right] \end{aligned}$$

$$\begin{aligned}
&= \frac{d}{dx} \left[\frac{63 \delta - 56 \delta}{72} \right] \\
&= \frac{d}{dx} \left[\frac{7}{72} \delta \right] \\
&\boxed{\frac{\tau_0}{\rho U^2} = \frac{7}{72} \frac{d\delta}{dx}} \\
\Rightarrow \quad \tau_0 &= \frac{7}{72} \rho U^2 \frac{d\delta}{dx}
\end{aligned}
\tag{2.17}$$

We know that,

$$\tau_0 = 0.0225 \rho U^2 \left(\frac{\mu}{\rho U \delta} \right)^{1/4} \tag{2.18}$$

Equating equation (2.17) and (2.18),

$$\begin{aligned}
\Rightarrow \quad \frac{7}{72} \rho U^2 \frac{d\delta}{dx} &= 0.0225 \rho U^2 \left(\frac{\mu}{\rho U \delta} \right)^{1/4} \\
\Rightarrow \quad \frac{7}{72} \frac{d\delta}{dx} &= 0.0225 \left(\frac{\mu}{\rho U \delta} \right)^{1/4} \\
\Rightarrow \quad d\delta &= 0.0225 \left(\frac{\mu}{\rho U \delta} \right)^{1/4} \times \frac{72}{7} dx \\
\Rightarrow \quad \delta^{1/4} d\delta &= 0.0225 \left(\frac{\mu}{\rho U} \right)^{1/4} \times \frac{72}{7} dx \\
\Rightarrow \quad \delta^{1/4} d\delta &= 0.2314 \left(\frac{\mu}{\rho U} \right)^{1/4} dx
\end{aligned}$$

Integrating

$$\begin{aligned}
\Rightarrow \quad \int \delta^{1/4} d\delta &= 0.2314 \left(\frac{\mu}{\rho U} \right)^{1/4} \int dx \\
\Rightarrow \quad \frac{\delta^{\frac{1}{4}+1}}{\frac{1}{4}+1} &= 0.2314 \left(\frac{\mu}{\rho U} \right)^{1/4} x + C
\end{aligned}$$

$$\Rightarrow \frac{\delta^{5/4}}{\frac{5}{4}} = 0.2314 \left(\frac{\mu}{\rho U} \right)^{1/4} x + C$$

$$\Rightarrow \frac{4}{5} \delta^{5/4} = 0.2314 \left(\frac{\mu}{\rho U} \right)^{1/4} x + C$$

Assuming boundary layer is turbulent over the entire length of the plate.

$$\text{So, at } x = 0, \delta = 0 \Rightarrow C = 0$$

$$\Rightarrow \frac{4}{5} \delta^{5/4} = 0.2314 \left(\frac{\mu}{\rho U} \right)^{1/4} x$$

$$\Rightarrow \delta^{5/4} = \frac{5}{4} \times 0.2314 \left(\frac{\mu}{\rho U} \right)^{1/4} x$$

$$\Rightarrow \delta = \left[\frac{5}{4} \times 0.2314 \left(\frac{\mu}{\rho U} \right)^{1/4} x \right]^{4/5}$$

$$= \left[0.289 \times \left(\frac{\mu}{\rho U} \right)^{1/4} x \right]^{4/5}$$

$$= (0.289)^{4/5} \times \left(\frac{\mu}{\rho U} \right)^{\frac{1}{4} \times \frac{4}{5}} \times x^{4/5}$$

$$= 0.370 \times \left(\frac{\mu}{\rho U x} \right)^{1/5} \times x^{1/5} \times x^{4/5}$$

$$= 0.370 \times \left(\frac{\mu}{\rho U x} \right)^{1/5} \times x$$

$$= 0.370 \times \left(\frac{\nu}{U x} \right)^{1/5} \times x \quad \left[\because \nu = \frac{\mu}{\rho} \right]$$

$$= 0.370 \times \left(\frac{1}{\text{Re}} \right)^{1/5} \times x \quad \left[\because \text{Re} = \frac{U x}{\nu} \right]$$

$$\delta = 0.370 (\text{Re})^{-0.2} \times x$$

Boundary layer thickness, $\delta = 0.370 (\text{Re})^{-0.2} \times x$
--

... (2.19)

Shear Stress, τ_0 :

We know that,

$$\text{Shear stress, } \tau_0 = 0.0225 \rho U^2 \left(\frac{\mu}{\rho U \delta} \right)^{1/4}$$

Substituting δ value,

$$\begin{aligned} \Rightarrow \tau_0 &= 0.0225 \rho U^2 \left[\frac{\mu}{\rho U \times 0.370 (\text{Re})^{-0.2} \times x} \right]^{1/4} \\ &= \frac{0.0225}{(0.370)^{1/4}} \rho U^2 \left[\frac{\mu}{\rho U x} \times (\text{Re})^{0.2} \right]^{1/4} \\ &= \frac{0.0225}{(0.370)^{1/4}} \rho U^2 \left[\frac{\nu}{U x} (\text{Re})^{0.2} \right]^{1/4} \left[\because \nu = \frac{\mu}{\rho} \right] \\ &= \frac{0.0225}{(0.370)^{1/4}} \rho U^2 \left[\frac{(\text{Re})^{0.2}}{\text{Re}} \right]^{1/4} \left[\because \text{Re} = \frac{U x}{\nu} \right] \\ &= \frac{0.0225}{(0.370)^{1/4}} \rho U^2 \left[(\text{Re})^{1/5} \times (\text{Re})^{-1} \right]^{1/4} \\ &= 0.02884 \rho U^2 [(\text{Re})^{-4/5}]^{1/4} \\ &= 0.02884 \rho U^2 [\text{Re}]^{-1/5} \\ &= \frac{0.05769}{2} \rho U^2 [\text{Re}]^{-0.2} \end{aligned}$$

$$\text{Shear stress, } \tau_0 = 0.05769 \frac{\rho U^2}{2} (\text{Re})^{-0.2}$$

Local Skin Friction Coefficient, C_{fx} :

We know that,

$$\text{Shear stress, } \tau_0 = 0.05769 \frac{\rho U^2}{2} (\text{Re})^{-0.2}$$

Also, we know $\tau_0 = C_{fx} \times \frac{\rho U^2}{2}$

Equating both equation,

$$\Rightarrow 0.05769 \frac{\rho U^2}{2} (\text{Re})^{-0.2} = C_{fx} \frac{\rho U^2}{2}$$

$$\Rightarrow C_{fx} = 0.05769 (\text{Re})^{-0.2}$$

$$\boxed{\text{Local friction coefficient, } C_{fx} = 0.05769 (\text{Re})^{-0.2}}$$

...(2.21)

Average friction coefficient,

We know that,

$$\begin{aligned} \text{Average friction coefficient, } C_{fL} &= \frac{1}{L} \int_0^L C_{fx} dx \\ &= \frac{1}{L} \int_0^L 0.05769 (\text{Re})^{-0.2} dx \\ &= \frac{1}{L} \int_0^L 0.05769 (\text{Re})^{-1/5} dx \\ &= \frac{1}{L} \int_0^L 0.05769 \left(\frac{Ux}{\nu} \right)^{-1/5} dx \\ &= \frac{1}{L} \times 0.05769 \times \left(\frac{U}{\nu} \right)^{-1/5} \int_0^L x^{-1/5} dx \\ &= \frac{1}{L} \times 0.05769 \times \left(\frac{U}{\nu} \right)^{-1/5} \left[\frac{x^{\frac{-1}{5}+1}}{\frac{-1}{5}+1} \right]_0^L \\ &= \frac{1}{L} \times 0.05769 \left(\frac{U}{\nu} \right)^{-1/5} \frac{L^{4/5}}{\frac{4}{5}} \end{aligned}$$

$$= \frac{5}{4} \times 0.05796 \left(\frac{U}{\nu} \right)^{-1/5} \times \frac{1}{L} \times L^{4/5}$$

$$= 0.072 \left(\frac{U}{\nu} \right)^{-1/5} \times L^{-1} \times L^{4/5}$$

$$= 0.072 \left(\frac{U}{\nu} \right)^{-1/5} \times L^{-1/5}$$

$$= 0.072 \left(\frac{UL}{\nu} \right)^{-1/5}$$

$$= 0.072 (Re)^{-1/5}$$

$$\boxed{\text{Average friction coefficient, } C_{fL} = 0.072 (Re)^{-0.2}} \dots (2.22)$$

$$\left[\because Re = \frac{UL}{\nu} \right]$$

1. Air at 20°C at atmospheric pressure flows over a flat plate at a velocity of 3 m/s. If the plate is 1 m wide and 80°C, calculate the following at $x = 300$ mm.

- 1. Hydrodynamic boundary layer thickness.**
- 2. Thermal boundary layer thickness,**
- 3. Local friction coefficient,**
- 4. Average friction coefficient,**
- 5. Local heat transfer coefficient,**
- 6. Average heat transfer coefficient,**
- 7. Heat transfer.**

Given:

Fluid temperature, $T_{\infty} = 20^{\circ}\text{C}$

Velocity, $U = 3$ m/s

Wide, $W = 1$ m

Surface temperature, $T_w = 80^{\circ}\text{C}$

Distance, $x = 300$ m = 0.3 m

To find:

1. Hydrodynamic boundary layer thickness,
2. Thermal boundary layer thickness,
3. Local friction coefficient,
4. Average friction coefficient,
5. Local heat transfer coefficient,
6. Average heat transfer coefficient,
7. Heat transfer.

Solution:

We know

$$\text{Film temperature } T_f = \frac{T_w + T_{\infty}}{2} = \frac{80 + 20}{2}$$

$T_f = 50^{\circ}\text{C}$

Properties of air at 50°C:

[From HMT data Book, Page No. 33]

Density, $\rho = 1.093 \text{ kg/m}^3$

Kinematic viscosity, $\nu = 17.95 \times 10^{-6} \text{ m}^2/\text{s}$

Prandtl Number = $\text{Pr} = 0.698$

Thermal conductivity, $k = 0.02826 \text{ W/mK}$

We know that,

$$\text{Reynolds Number, } \text{Re} = \frac{UL}{\nu}$$

$$\frac{3 \times 0.3}{17.95 \times 10^{-6}} \quad [\because x = L = 0.3\text{m}]$$

$$\boxed{\text{Re} = 5.01 \times 10^4} < 5 \times 10^5$$

Since $\text{Re} < 5 \times 10^5$, flow is laminar

For Flat plate, laminar flow,

[Refer HMT data Book, Page No. 112]

1. Hydrodynamic boundary layer thickness:

$$\begin{aligned}\delta_{hx} &= 5 \times x \times (\text{Re})^{-0.5} \\ &= 5 \times 0.3 \times (5.01 \times 10^4)^{-0.5}\end{aligned}$$

$$\boxed{\delta_{hx} = 6.7 \times 10^{-3} \text{ m}}$$

2. Thermal boundary layer thickness:

$$\begin{aligned}\delta_{Tx} &= \delta_{hx} (\text{Pr})^{-0.333} \\ \delta_{hx} &= (6.7 \times 10^{-3}) (0.698)^{-0.333}\end{aligned}$$

$$\boxed{\delta_{hx} = 7.5 \times 10^{-3} \text{ m}}$$

3. Local Friction coefficient:

$$C_{fx} = 0.664(\text{Re})^{-0.5}$$
$$= 0.664(5.01 \times 10^4)^{-0.5}$$

$$C_{fx} = 2.96 \times 10^{-3}$$

4. Average friction coefficient:

$$\overline{C_{fL}} = 1.328(\text{Re})^{-0.5}$$
$$= 1.328(5.01 \times 10^4)^{-0.5}$$
$$= 5.9 \times 10^{-3}$$

$$\overline{C_{fL}} = 5.9 \times 10^{-3}$$

5. Local heat transfer coefficient (h_x):

Local Nusselt Number

$$\text{Nu}_x = 0.332 (\text{Re})^{0.5} (\text{Pr})^{0.333}$$
$$= 0.332 (5.01 \times 10^4)^{0.5} (0.698)^{0.333}$$

$$\text{Nu}_x = 65.9$$

We know,

$$\text{Nu}_x = \frac{h_x \times L}{k}$$

$$65.9 = \frac{h_x \times 0.3}{0.02826} \quad [\because x = L = 0.3\text{m}]$$

$$h_x = 6.20 \text{ W/m}^2 \text{ K}$$

6. Average heat transfer coefficient (h):

$$h = 2 \times h_x$$

$$= 2 \times 6.20$$

$$h = 12.41 \text{ W/m}^2 \text{ K}$$

7. Heat transfer:

We know that,

$$\begin{aligned} Q &= hA(T_w - T_\infty) \\ &= 12.41 \times (1 \times 0.3)(80 - 20) \end{aligned}$$

$Q = 223.38$

Result:

1. $\delta_{hx} = 6.7 \times 10^{-3} \text{ m}$,
2. $\delta_{Tx} = 7.5 \times 10^{-3} \text{ m}$,
3. $C_{fx} = 2.96 \times 10^{-3}$,
4. $\overline{C_{fL}} = 5.9 \times 10^{-3}$,
5. $h_x = 6.20 \text{ W / m}^2 \text{ K}$
6. $h = 12.41 \text{ W/m}^2\text{K}$,
7. $Q = 223.38 \text{ W}$.

2. Air at 30°C flow over a flat plate at a velocity of 2 m/s. The plate is 2 m long and 1.5 m wide. Calculate the following:

1. Hydrodynamic and thermal boundary layer thickness at the trailing edge of the plate,
2. Total drag force,
3. Total mass flow rate through the boundary layer between $x = 40 \text{ cm}$ and $x = 85 \text{ cm}$.

Given:

Fluid temperature, $T_\infty = 30^\circ\text{C}$

Velocity, $U = 2 \text{ m/s}$

Length, $L = 2 \text{ m}$

Wide, $W = 1.5 \text{ m}$

To find:

1. Hydrodynamic and thermal boundary layer thickness.
2. Total drag force.
3. Total mass flow rate through the boundary layer between $x = 40 \text{ cm}$ and $x = 85 \text{ cm}$.

Solution:

Properties of air at 30°C

[From HMT data Book, Page No. 33]

$$\rho = 1.165 \text{ kg/m}^3$$

$$\nu = 16 \times 10^{-6} \text{ m}^2/\text{s}$$

$$\text{Pr} = 0.701$$

$$K = 0.02675 \text{ W/mK}$$

We know that,

$$\begin{aligned} \text{Reynolds Number, Re} &= \frac{UL}{\nu} \\ &= \frac{2 \times 2}{16 \times 10^{-6}} \\ \boxed{\text{Re} = 2.5 \times 10^5} &< 5 \times 10^5 \end{aligned}$$

Since $\text{Re} < 5 \times 10^5$, flow is laminar

For flat plate, laminar flow,

[From HMT data Book, Page No. 112]

Hydrodynamic boundary layer thickness

$$\begin{aligned} \delta_{Tx} &= 5 \times x \times (\text{Re})^{-0.5} \\ &= 5 \times 2 \times (2.5 \times 10^5)^{-0.5} \end{aligned} \quad [\because x = L = 2\text{m}]$$

$$\boxed{\delta_{Tx} = 0.02\text{m}}$$

Thermal boundary layer thickness,

$$\begin{aligned} \delta_{Tx} &= \delta_{hx} \times (\text{Pr})^{-0.333} \\ &= 0.02 \times (0.701)^{-0.333} \end{aligned}$$

$$\boxed{\delta_{Tx} = 0.02}$$

Average friction coefficient,

$$\overline{C_{fL}} = 1.328(\text{Re})^{-0.5}$$

$$= 1.328 \times (2.5 \times 10^5)^{-0.5}$$

$$C_{fL} = 1.328 (\text{Re})^{-0.5}$$

$$= 1.328 \times (2.5 \times 10^5)^{-0.5}$$

$$C_{fL} = \frac{\tau}{\frac{\rho U^2}{2}}$$

$$\Rightarrow 2.65 \times 10^{-3} = \frac{\tau}{\frac{1.165 \times (2)^2}{2}}$$

$$\Rightarrow \boxed{\text{Average shear stress, } \tau = 6.1 \times 10^{-3} \text{ N/m}^2}$$

Drag force = Area $\times$ Average shear stress

$$= 2 \times 1.5 \times 6.1 \times 10^{-3}$$

$$\boxed{\text{Drag force} = 0.018}$$

Drag force on two sides of the plate

$$= 0.018 \times 2 = 0.036 \text{ N}$$

$$\boxed{\text{Drag force, } F_D = 0.036 \text{ N}} \quad x = 50$$

Total mass flow rate between $x = 40$ cm and

..... (1)

$$\Delta m = \frac{5}{8} \rho U \left[\delta \right]_{hx=0}^{hx=40}$$

Hydrodynamic boundary layer thickness

$$\begin{aligned}
\delta_{hx} &= 0.85 = 5 \times x \times (\text{Re})^{-0.5} \\
&= 5 \times 0.85 \times \left[\frac{U \times x}{\nu} \right]^{-0.5} \\
&= 5 \times 0.85 \times \left[\frac{2 \times 0.85}{16 \times 10^{-6}} \right]^{-0.5} \\
&\quad [\because x = 85 \text{ cm} = 0.85 \text{ m}]
\end{aligned}$$

$$\delta_{hx=0.85} = 0.0130 \text{ m}$$

$$\begin{aligned}
\delta_{hx} &= 0.40 = 5 \times x \times (\text{Re})^{-0.5} \\
&= 5 \times 0.40 \times \left(\frac{2 \times 0.40}{16 \times 10^{-6}} \right)^{-0.5}
\end{aligned}$$

$$\delta_{hx=0.40} = 8.9 \times 10^{-3} \text{ m}$$

$$(1) \Rightarrow \Delta m = \frac{5}{8} \times 1.165 \times 2 [0.0130 - 8.9 \times 10^{-3}]$$

$$\Delta m = 5.97 \times 10^{-3} \text{ kg / s}$$

Result:

1. Hydrodynamic boundary layer thickness, $\delta_{hx} = 0.02 \text{ m}$,
Thermal boundary layer thickness, $\delta_{Tx} = 0.0225 \text{ m}$,
2. Drag force, $F_D = 0.036 \text{ N}$,
3. Total mass flow rate, $\Delta m = 5.97 \times 10^{-3} \text{ kg/s}$

3. Air at 30°C flows over a flat plate at a velocity of 4 m/s . The plate is maintained at 90°C . The plate dimension is $90 \times 30 \text{ cm}^2$. Calculate the heat transfer for the following condition

1. First half of the plate,
2. Full plate,
3. Next half of the plate.

Given:

Fluid temperature, $T_{\infty} = 30^{\circ}\text{C}$

Velocity, $U = 4 \text{ m/s}$

Plate surface temperature, $T_w = 90^{\circ}\text{C}$

Plate dimension = $90 \times 30 \text{ cm}^2$
 $= 0.90 \times 0.30 \text{ m}^2$

To find:

Heat transfer for

1. First half of the plate, i.e., $x = 0.45 \text{ m}$,
2. Full plate, i. e., $x = 0.90 \text{ m}$,
3. Next half of the plate.

Solution:

We know that,

$$\begin{aligned} \text{Film temperature, } T_f &= \frac{T_w + T_{\infty}}{2} \\ &= \frac{90 + 30}{2} \end{aligned}$$

$T_f = 60^{\circ}\text{C}$

Properties of air at 60°C :

[From HMT data Book, Page No. 33]

$$\rho = 1.060 \text{ kg/m}^3$$

$$\nu = 18.97 \times 10^{-6} \text{ m}^2/\text{s}$$

$$\text{Pr} = 0.696$$

$$K = 0.02896 \text{ W/mK}$$

$$\text{Reynolds number, } \text{Re} = \frac{U \times L}{\nu}$$

$$\frac{4 \times 0.45}{18.97 \times 10^{-6}}$$

$$\text{Re} = 9.4 \times 10^4 < 5 \times 10^5$$

Since $\text{Re} < 5 \times 10^5$, flow is laminar

[From HMT data Book, Page No. 112]

$$\text{Local Nusselt Number, } \text{Nu}_x = \frac{h_x \times L}{k}$$

$$90.21 = \frac{h_x \times 0.45}{0.02896}$$

==>

$$h_x = 5.80 \text{ W/m}^2\text{K}$$

$$\text{Local heat transfer coefficient, } h_x = 5.80 \text{ W/m}^2\text{K}$$

Average heat transfer coefficient

$$h = 2 \times h_x$$

$$h = 11.61 \text{ W/m}^2\text{K}$$

$$\text{Heat transfer, } Q_1 = h \times A \times (T_w - T_\infty)$$

$$= h \times L \times W \times (T_w - T_\infty)$$

$$= 11.61 \times 0.45 \times 0.30 \times (90 - 30)$$

$$[\because x = L = 0.45\text{m}; W = 0.30\text{m}]$$

$$Q_1 = 94.04 \text{ W}$$

Case (ii):

For full pate $x = L = 0.9 \text{ m}$

$$\text{Reynolds Number, Re} = \frac{U \times L}{\nu}$$

$$\frac{4 \times 0.90}{18.97 \times 10^{-6}}$$

$$\boxed{\text{Re} = 1.89 \times 10^5 < 5 \times 10^5}$$

Since $\text{Re} < 5 \times 10^5$, flow is laminar

For flat plate, laminar flow,

$$\text{Local Nusselt Number, Nu}_x = 0.332(\text{Re})^{0.5} (\text{Pr})^{0.333}$$

$$= 0.332 (1.89 \times 10^5)^{0.5} \times (0.696)^{0.333}$$

$$\boxed{\text{Nu}_x = 128.18}$$

We know,

$$\text{Nu}_x = \frac{h_x \times L}{k}$$

$$128.18 \frac{h_x \times 0.90}{0.02896}$$

Local heat transfer coefficient

$$h_x = 4.12 \text{ W} / \text{m}^2 \text{ K}$$

Average heat transfer coefficient

$$h = 2 \times h_x = 2 \times 4.12$$

$$\boxed{h = 8.24 \text{ W/m}^2 \text{ K}}$$

Heat transfer for entire plate

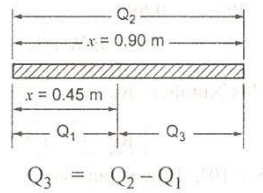
$$Q_2 = h \times A \times (T_w - T_\infty)$$

$$= 8.24 \times 0.90 \times 0.30 \times (90 - 30)$$

$$\boxed{Q_2 = 133.48 \text{ W}}$$

Case (iii):

Heat lost from the next half of the plate



$$Q_3 = Q_2 - Q_1$$
$$= 133.48 - 94.04$$

$$Q_3 = 39.44 \text{ W}$$

Result:

- | | |
|--|--------------------------|
| 1. Heat lost for first half of the plate | $Q_1 = 94.04 \text{ W}$ |
| 2. Heat lost for entire plate | $Q_2 = 133.48 \text{ W}$ |
| 3. Heat lost for next half of the plate | $Q_3 = 39.44 \text{ W}$ |

4. Air at 40°C flows over a flat plate, 0.8 m long at a velocity of 50 m/s . The plate surface is maintained at 300°C . Determine the heat transferred from the entire plate length to air taking into consideration both laminar and turbulent portion of the boundary layer. Also calculate the percentage error if the boundary layer is assumed to be turbulent nature from the very leading edge of the plate.

Given:

Fluid temperature, $T_\infty = 40^\circ\text{C}$

Length, $L = 0.8 \text{ m}$

Velocity, $U = 50 \text{ m/s}$

Plate surface temperature, $T_w = 300^\circ\text{C}$

To find:

1. Heat transferred for:
 - (i) Entire plate is considered as combination of both laminar and turbulent flow.
 - (ii) Entire plate is considered as turbulent flow.

2. Percentage error.

Solution:

We know that,

$$\begin{aligned}\text{Film temperature, } T_f &= \frac{T_w + T_\infty}{2} \\ &= \frac{300 + 40}{2} = 443K\end{aligned}$$

$$T_f = 170^\circ\text{C}$$

Properties of air at 170°C :

$$\rho = 0.790 \text{ kg/m}^3$$

$$\nu = 31.10 \times 10^{-6} \text{ m}^2/\text{s}$$

$$\text{Pr} = 0.6815$$

$$k = 0.037 \text{ W/mK}$$

We know

$$\begin{aligned}\text{Reynolds Number, } Re &= \frac{UL}{\nu} \\ &= \frac{50 \times 0.8}{31.10 \times 10^{-6}} = 1.26 \times 10^6\end{aligned}$$

$$Re = 1.286 \times 10^6 > 5 \times 10^5$$

$Re > 5 \times 10^5$, so this is turbulent flow.

Case (i):

Laminar-turbulent combined. [It means, flow is laminar upto Reynolds number value is 5×10^5 , after that flow is turbine.]

$$\text{Average Nusselt Number } Nu = (\text{Pr})^{0.333} [0.037 (Re)^{0.8} - 871]$$

[From HMT data Book, Page No. 114]

$$Nu = (0.6815)^{0.333} [0.037(1.286 \times 10^6)^{0.8} - 871]$$

$$\text{Average Nusselt Number } Nu = 1746.09$$

We know

$$Nu = \frac{hL}{k}$$

$$1746.09 = \frac{h \times 0.8}{0.037}$$

$$h = 80.75 \text{ W/m}^2\text{K}$$

Average heat transfer coefficient $h = 80.75 \text{ W/m}^2\text{K}$

Heat transfer, $Q_1 = h \times A \times (T_w - T_\infty)$

$$\begin{aligned} & h \times L \times W \times (T_w - T_\infty) \\ &= h \times L \times W (T_w - T_\infty) \\ &= 80.75 \times 0.8 \times 1 \times (300 - 40) \end{aligned}$$

$$Q_1 = 16796$$

Case (ii):

Entire plate is turbulent flow:

$$\text{Local Nusselt Number} \quad Nu_x = 0.029 \times (Re)^{0.8} \times (Pr)^{0.33}$$

[From HMT data Book, Page No. 113]

$$Nu_x = 0.029 \times (1.286 \times 10^6)^{0.8} \times (0.6815)^{0.33}$$

$$Nu_x = 2010.15$$

We know

$$Nu_x = \frac{h_x \times L}{k}$$

$$2010.15 = \frac{h_x \times 0.8}{0.037}$$

$$h_x = 92.96 \text{ W/m}^2\text{K}$$

Local heat transfer coefficient, $h_x = 92.96 \text{ W/m}^2\text{K}$

Average heat transfer coefficient (for fully turbulent flow) $h = 1.25 \times h_x$

$$= 1.25 \times 92.96$$

Average heat transfer coefficient $h = 116.20 \text{ W/m}^2\text{K}$

$$\begin{aligned}\text{Heat transfer, } Q_2 &= h \times A \times W \times (T_w - T_\infty) \\ &= h \times L \times W (T_w - T_\infty) \\ &= 116.20 \times 0.8 \times 1 \times (300 - 40)\end{aligned}$$

$$Q_2 = 24169.60 \text{ W}$$

$$\begin{aligned}\text{Percentage error} &= \frac{Q_2 - Q_1}{Q_1} \\ &= \frac{24169.60 - 16796}{16796} \times 100 \\ &= 43.90\end{aligned}$$

Result:

1. Heat transfer (Laminar – Turbulent combined) $Q_1 = 16796 \text{ W}$
2. Heat transfer (Fully turbulent) $Q_2 = 24169.60 \text{ W}$
3. Percentage error 43.90

5. Air at 15°C , 30 Km/h flows over a cylinder of 400 mm diameter and 1500 mm height with surface temperature of 45°C . Calculate the heat loss.

Given:

Fluid temperature,

$$T_\infty = 15^\circ\text{C}$$

Velocity, $U = 30 \text{ km/h}$

$$U = \frac{30 \times 10^3 \text{ m}}{3600 \text{ s}}$$

$$U = 8.33 \text{ m/s}$$

$$\text{Diameter, } D = 400 \text{ mm} = 0.4 \text{ m}$$

$$\text{Length, } L = 1500 \text{ mm} = 1.5 \text{ m}$$

$$\text{Plate surface temperature, } T_w = 45^\circ\text{C}$$

To find:

Heat loss

Solution:

We know that,

$$\text{Film temperature, } T_f = \frac{T_w + T_\infty}{2} = \frac{45 + 15}{2}$$

$$T_f = 30^\circ\text{C}$$

Properties of air at 30°C :

[From HMT data Book, Page No. 33]

$$\text{Density, } \rho = 1.165 \text{ kg/m}^3$$

$$\text{Kinematic viscosity, } \nu = 16 \times 10^{-6} \text{ m}^2/\text{s}$$

$$\text{Prandtl Number, } Pr = 0.701$$

$$\text{Thermal conductivity, } k = 0.02675 \text{ W/mK}$$

We know,

$$\begin{aligned} \text{Reynolds Number, } Re &= \frac{UD}{\nu} \\ &= \frac{8.33 \times 0.4}{16 \times 10^{-6}} \end{aligned}$$

$$Re_D = 2.08 \times 10^5$$

$$\text{Nusselt Number, } Nu = C (Re)^m (Pr)^{0.333}$$

[From HMT data Book, Page No. 115]

Re_D value is 2.08×10^5 , corresponding C value is 0.0266 and m value is 0.805.

$$\Rightarrow Nu = 0.0266 \times (2.08 \times 10^5)^{0.805} \times (0.701)^{0.333}$$

$$Nu = 451.3$$

We know that,

$$\text{Nusselt number, } Nu = \frac{hD}{k}$$

$$\Rightarrow 451.3 = \frac{h \times 0.4}{0.02675}$$

$$\Rightarrow h = 30.18 \text{ W/m}^2\text{K}$$

$$\text{Heat transfer coefficient, } h = 30.18$$

$$\text{Heat transfer, } Q = hA(T_w - T_\infty)$$

$$= h \times \pi \times D \times L \times (T_w - T_\infty)$$

$$[\because A = \pi DL]$$

$$= 30.18 \times \pi \times 0.4 \times 1.5 \times (45 - 15)$$

$$Q = 1706.6 \text{ W}$$

Result:

Heat loss, $Q = 1706.6 \text{ W}$

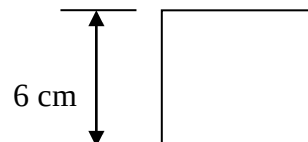
6. Air at 40°C flows over a tube with a velocity of 30 m/s . The tube surface temperature is 120°C ; calculate the heat transfer coefficient for the following cases.

- 1. Tube could be square with a side of 6 cm .**
- 2. Tube is circular cylinder of diameter 6 cm .**

Given:

Fluid temperature, $T_\infty = 40^\circ\text{C}$

Velocity, $U = 30 \text{ m/s}$



Tube surface temperature, $T_w = 120^\circ\text{C}$

To find:

Heat transfer coefficient, (h)

Solution:

We know that,

$$\begin{aligned}\text{Film, temperature, } T_f &= \frac{T_w + T_\infty}{2} \\ &= \frac{120 + 40}{2}\end{aligned}$$

$$T_f = 80^\circ\text{C}$$

Properties of air at 80°C

[From HMT data Book, Page No. 33]

$$\rho = 1\text{kg/m}^3$$

$$\nu = 21.09 \times 10^{-6} \text{ m}^2/\text{s}$$

$$\text{Pr} = 0.692$$

$$k = 0.03047 \text{ W/mK}$$

Case (i):

Tube is considered as square of side 6 cm.

$$\text{i.e., } L = 6 \text{ cm} = 0.06 \text{ m}$$

$$\begin{aligned}\text{Reynolds Number, } Re &= \frac{UD}{\nu} \\ &= \frac{30 \times 0.06}{21.09 \times 10^{-6}}\end{aligned}$$

$$Re = 0.853 \times 10^5$$

$$\text{Nusselt Number, } Nu = C \times (Re)^n (Pr)^{0.333}$$

For square, $n = 0.675$

$$C = 0.092$$

[From HMT data Book, Page No. 118]

$$\Rightarrow Nu = 0.092 (0.853 \times 10^5)^{0.675} \times (0.692)^{0.333}$$

$\Rightarrow$

$$Nu = 173.3$$

We know that,

$$Nu = \frac{hL}{k}$$

$$173.3 = \frac{h \times 0.06}{0.03047}$$

$$\text{Heat transfer coefficient, } h = 88$$

Case (ii):

Tube diameter, $D = 6 \text{ cm} = 0.06 \text{ m}$

$$\text{Reynolds Number, } Re = \frac{UD}{\nu}$$

$$= \frac{30 \times 0.06}{21.09 \times 10^{-6}}$$

$$Re_D = 0.853 \times 10^5$$

$$\text{Nusselt Number, } Nu = C (Re)^m (Pr)^{0.333}$$

[From HMT data Book, Page No. 115]

Re_D value is 0.853×10^5 , corresponding C and m values are 0.0266 and 0.805 respectively.

$$\Rightarrow Nu = 0.0266 \times (0.853 \times 10^5)^{0.805} \times (0.692)^{0.333}$$

$$Nu = 219.3$$

We know that,

$$\text{Nusselt number, } Nu = \frac{hD}{k}$$

$$\Rightarrow 219.3 = \frac{h \times 0.06}{0.03047}$$

$$\Rightarrow h = 111.3 \text{ W/m}^2\text{K}$$

$$\text{Heat transfer coefficient, } h = 111.3 \text{ W/m}^2\text{K}$$

Result:

1. Heat transfer coefficient for square tube $h = 88 \text{ W/m}^2\text{K}$
2. Heat transfer coefficient for circular tube $h = 111.3 \text{ W/m}^2\text{K}$

7. in a surface condenser, water flows through staggered tubes while the air is passed in cross flow over the tubes. The temperature and velocity of air are 30°C and 8 m/s respectively. The longitudinal and transverse pitches are 22 mm and 20 mm respectively. The tube outside diameter is 18 mm and tube surface temperature is 90°C . Calculate the heat transfer coefficient.

Given:

Fluid temperature, $T_\infty = 30^\circ\text{C}$

Velocity, $U = 8 \text{ m/s}$

Longitudinal pitch, $S_t = 22 \text{ mm} = 0.022 \text{ m}$

Transverse pitch, $S_t = 20 \text{ mm} = 0.020 \text{ m}$

Diameter, $D = 18 \text{ mm} = 0.018 \text{ m}$

Tube surface temperature, $T_w = 90^\circ\text{C}$

To find:

1. Heat transfer coefficient

Solution:

We know that,

$$\begin{aligned}\text{Film, temperature, } T_f &= \frac{T_w + T_\infty}{2} \\ &= \frac{90 + 30}{2}\end{aligned}$$

$$T_f = 60^\circ\text{C}$$

Properties of air at 60°C

[From HMT data Book, Page No. 33]

$$\rho = 1.060\text{kg/m}^3$$

$$\nu = 18.97 \times 10^{-6} \text{ m}^2/\text{s}$$

$$\text{Pr} = 0.692$$

$$k = 0.02896 \text{ W/mK}$$

We know that,

$$\text{Maximum Velocity, } U_{\max} = U \times \frac{S_t}{S_t - D}$$

$$U_{\max} = 8 \times \frac{0.020}{0.020 - 0.018}$$

$$U_{\max} = 80 \text{ m/s}$$

$$\text{Reynolds Number, } \text{Re} = \frac{U_{\max} \times D}{\nu}$$

$$= \frac{80 \times 0.018}{18.97 \times 10^{-6}}$$

$$\text{Re} = 7.5 \times 10^4$$

$$\text{Nusselt Number, } Nu = C \times (Re)^n (Pr)^{0.333}$$

$$\frac{S_t}{D} = \frac{0.020}{0.018} = 1.11$$

$$\boxed{\frac{S_t}{D} = 1.11}$$

$$\frac{S_t}{D} = \frac{0.022}{0.018} = 1.22$$

$$\boxed{\frac{S_t}{D} = 1.22}$$

$$\frac{S_t}{D} = 1.11, \frac{S_l}{D} = 1.22, \text{ Corresponding } C, n \text{ values are } 0.518 \text{ and } 0.556 \text{ respectively.}$$

[From HMT data Book, Page No. 112]

$$\boxed{C = 0.518}$$

$$\boxed{n = 0.556}$$

We know that,

$$\text{Nusselt Number, } Nu = 1.13 (Pr)^{0.333} [C (Re)^n]$$

[From HMT data Book, Page No. 112]

$$\Rightarrow Nu = 1.13 \times (0.693)^{0.333} \times 10^5 \times [0.518 \times (7.5 \times 10^4)^{0.556}]$$

$$\boxed{Nu = 266.3}$$

$$173.3 = \frac{h \times 0.06}{0.03047}$$

$$\Rightarrow Nu = 0.0266 \times (0.853 \times 10^5)^{0.805} \times (0.692)^{0.333}$$

$$\boxed{Nu = 219.3}$$

$$\text{Nusselt number, } Nu = \frac{hD}{k}$$

$$\Rightarrow 266.3 = \frac{h \times 0.018}{28.96 \times 10^{-3}}$$

Heat transfer coefficient, $h = 428.6 \text{ W/m}^2\text{K}$
--

Result:

Heat transfer coefficient $h = 4238.6 \text{ W/m}^2\text{K}$

8. Water at 50°C enters 50 mm diameter and 4 m long tube with a velocity of 0.8 m/s. The tube wall is maintained at a constant temperature of 90°C . Determine the heat transfer coefficient and the total amount of heat transferred if exit water temperature is 70°C .

Given:

Inner temperature of water, $T_{mi} = 50^\circ\text{C}$

Diameter, $D = 50 \text{ mm} = 0.05 \text{ m}$

Length, $L = 4 \text{ m}$

Velocity, $U = 0.8 \text{ m/s}$

Tube wall temperature, $T_w = 90^\circ\text{C}$

Exit temperature of water $T_{mo} = 70^\circ\text{C}$

To find:

1. Heat transfer coefficient, (h)
2. Heat transfer, (Q)

Solution:

Bulk mean temperature, $T_m = \frac{T_{mi} + T_{mo}}{2}$

$$= \frac{50 + 70}{2}$$

$T_m = 60^\circ\text{C}$

Properties of air at 60°C

[From HMT data Book, Page No. 21]

$$\rho = 985 \text{ kg/m}^3$$

$$\nu = 0.478 \times 10^{-6} \text{ m}^2/\text{s}$$

$$\text{Pr} = 3.020$$

$$k = 0.6513 \text{ W/mK}$$

let us first determine the type of flow:

$$\begin{aligned} \text{Re} &= \frac{UD}{\nu} \\ &= \frac{0.8 \times 0.05}{0.478 \times 10^{-6}} \end{aligned}$$

$$\text{Re} = 8.36 \times 10^4$$

Since $\text{Re} > 2300$, flow is turbulent.

$$\frac{L}{D} = \frac{4}{0.05} = 80$$

$$\frac{L}{D} = 80 > 60$$

$$\text{Re} = 8.36 \times 10^4 > 10,000$$

$$\text{Pr} = 3.020 \implies 0.6 < \text{Pr} < 160$$

$\frac{L}{D}$ Ratio is greater than 60. Re value is greater than 10,000 and Pr value is in between

0.6 and 160. So,

$$\text{Nusselt Number, Nu} = 0.023 (\text{Re})^{0.8} (\text{Pr})^n$$

[From HMT data Book, Page No. 125]

[Inlet temperature 50°C, Exit temperature 70°C → Heating Process, So, $n = 0.4$]

$$\text{Nu} = 0.023 \times (8.36 \times 10^4)^{0.8} \times (3.020)^{0.4}$$

$$\text{Nu} = 310$$

We know that, $Nu = \frac{hD}{k}$

$$310 = \frac{h \times 0.05}{0.6513}$$

Heat transfer coefficient, $h = 4039.3$

$$\text{Heat transfer, } Q = hA(T_w - T_\infty)$$

$$= h \times \pi \times D \times L \times (T_w - T_\infty)$$

$$= 4039.3 \times \pi \times D \times L \times (T_w - T_\infty)$$

$Q = 76139 \text{ W}$

Result:

1. Heat transfer coefficient $h = 4039.3 \text{ W/m}^2\text{K}$
2. Heat transfer, $Q = 76139 \text{ W}$.

9. Air at 30°C , 6 m/s flows in a rectangular section of size $300 \times 800 \text{ mm}$. calculate the heat leakage per metre length per unit temperature difference.

Given:

$$\text{Air temperature, } T_m = 30^\circ\text{C}$$

$$\text{Velocity, } U = 6 \text{ m/s}$$

$$\text{Area, } A = 300 \times 800 \text{ mm}^2$$

$$= 0.3 \times 0.8 \text{ m}^2$$

To find:

1. Heat leakage per metre length per unit temperature difference.

Solution:

Properties of air at 30°C:

$$\rho = 1.165 \text{ kg/m}^3$$

$$\nu = 16 \times 10^{-6} \text{ m}^2/\text{s}$$

$$\text{Pr} = 0.701$$

$$k = 0.02675 \text{ W/mK}$$

Equivalent diameter for $300 \times 800 \text{ mm}^2$ cross section is given by

$$D_e = \frac{4A}{P} = \frac{4 \times (0.3 \times 0.8)}{2(0.3 + 0.8)}$$

Where

$$P - \text{Perimeter} = 2(L + W)$$

$$D_e = 0.436 \text{ m}$$

We know that,

$$\text{Reynolds Number, } \text{Re} = \frac{UD_e}{\nu}$$

$$= \frac{6 \times 0.436}{16 \times 10^{-6}}$$

$$\text{Re} = 16.3 \times 10^4$$

Since $\text{Re} > 2300$, flow is turbulent.

For turbulent flow general equation is ($\text{Re} > 10000$),

$$\text{Nu} = 0.023(\text{Re})^{0.8} (\text{Pr})^n$$

[From HMT data Book, Page No. 125]

Assuming the pipe wall temperature to be higher than air temperature. So, heating process ==> $n = 0.4$.

$$\text{==> } \text{Nu} = 0.023 \times (16.3 \times 10^4)^{0.8} \times (0.701)^{0.4}$$

$$\text{Nu} = 294.96$$

We know,

$$\text{Nusselt Number, } Nu = \frac{hD_e}{k}$$

$$294.96 = \frac{h \times 0.436}{26.75 \times 10^{-3}}$$

Heat transfer coefficient, $h = 18.09 \text{ W/m}^2\text{K}$
--

Heat leakage per unit length per unit temperature difference

$$\begin{aligned} Q &= hP \\ &= 18.09[2 \times (0.3 + 0.8)] \end{aligned}$$

$Q = 39.79 \text{ W}$

Result:

Heat leakage, $Q = 39.79 \text{ W}$.

10. Air at 2 bar pressure and 60°C is heated as it flows through a tube of diameter 25 mm at a velocity of 15 m/s. If the wall temperature is maintained at 100°C , find the heat transfer per unit length of the tube. How much would be the bulk temperature increase over one metre length of the tube.

Given:

Pressure, $P = 2 \text{ bar} = 2 \times 10^5 \text{ N/m}^2$

Inlet temperature of air, $T_{mi} = 60^\circ\text{C}$

Diameter of tube $D = 25 \text{ mm}^2 = 0.05 \text{ m}$

Velocity, $U = 15 \text{ m/s}$

Tube wall temperature, $T_w = 100^\circ\text{C}$

Length, $L = 1 \text{ m}$

To find:

1. Heat transfer per unit length of the tube, Q.
2. Rise in bulk temperature of air, $(T_{mo} - T_{mi})$

Solution:

Properties of air at 60°C:

[From HMT data Book, Page No. 33]

$$\rho = 1.060 \text{ kg/m}^3$$

$$\nu = 18.97 \times 10^{-6} \text{ m}^2/\text{s}$$

$$\text{Pr} = 0.696$$

$$k = 0.02896 \text{ W/mK}$$

Note:

Given pressure is above atmospheric pressure. So, kinematic viscosity, ν and density, ρ will vary with pressure. Pr, k , C_p are same for all pressures.

$$\text{Kinematic viscosity, } \nu = \nu_{atm} \times \frac{P_{atm}}{P_{given}}$$

$$= 18.97 \times 10^{-6} \times \frac{1 \text{ bar}}{2 \text{ bar}}$$

[$\because$ Atmospheric pressure = 1 bar]

$$= 18.97 \times 10^{-6} \times \frac{1 \times 10^5}{2 \times 10^5}$$

$$\boxed{\nu = 9.485 \times 10^{-6} \text{ m}^2/\text{s}}$$

$$\text{Density, } \rho = \frac{P}{RT}$$

$$= \frac{2 \times 10^5}{287 \times (60 + 273)}$$

$$\boxed{\rho = 2.092 \text{ kg/m}^3}$$

We know that,

$$\text{Reynolds Number, Re} = \frac{UD}{\nu}$$
$$= \frac{15 \times 0.025}{9.485 \times 10^{-6}}$$



$$\text{Re} = 39.53 \times 10^3$$

2300

Since $\text{Re} > 2300$, flow is turbulent.

For turbulent flow general equation is ($\text{Re} > 10000$),

$$\text{Nu} = 0.023(\text{Re})^{0.8} (\text{Pr})^n$$

[From HMT data Book, Page No. 125]

This is heating process. So, $n = 0.4$.

$$\Rightarrow \text{Nu} = 0.023 \times (39.53 \times 10^3)^{0.8} \times (0.696)^{0.4}$$

$$\text{Nu} = 94.70$$

We know,

$$\text{Nu} = \frac{hD}{k}$$

$$94.70 = \frac{h \times 0.025}{0.02896}$$

$$\Rightarrow h = 109.70 \text{ W/m}^2\text{K}$$

$$\text{Heat transfer coefficient, } h = 18.09 \text{ W/m}^2\text{K}$$

Mass flow rate, $m = \rho AU$

$$= \rho \times \frac{\pi}{4} D^2 \times U$$
$$= 2.092 \times \frac{\pi}{4} (0.025)^2 \times 15$$

$$m = 0.015 \text{ kg/s}$$

We know that,

$$\text{Heat transfer, } Q = mC_p(T_{mo} - T_{mi})$$

$$= 0.015 \times 1005(T_{mo} - 60)$$

$$[\because \text{For air } C_p = 1005 \text{ J/kgK}]$$

Q = Heat leakage per unit length per unit temperature difference

$$Q = hP(T_{mo} - 60) \quad \dots\dots (1)$$

We know that,

$$\text{Heater transfer, } Q = mC_p(T_{mo} - T_{mi})$$

$$= h \times \pi DL \times (T_w - T_m)$$

$$= 109.70 \times \pi 0.025 \times 1 \times (100 - T_m) \quad \dots\dots (2)$$

Equating (1) and (2),

$$15.075(T_{mo} - 60) = 8.615(100 - T_m)$$

$$\Rightarrow 1.749(T_{mo} - 60) = 100 - T_m$$

$$\Rightarrow 1.749(T_{mo} - 60) = 100 - \frac{T_{mi} + T_{mo}}{2}$$

$$\Rightarrow 1.749T_{mo} - 104.94 = 100 - \frac{T_{mo}}{2}$$

$$\Rightarrow 1.749T_{mo} - 104.94 = 100 - 30 - \frac{T_{mo}}{2}$$

$$\Rightarrow 1.749T_{mo} + \frac{T_{mo}}{2} = 100 - 30 + 104.94$$

$$\Rightarrow 2.249T_{mo} = 174.94$$

$$\Rightarrow T_{mo} = 77.78^\circ \text{C}$$

Outlet temperature of air, $T_{mo} = 77.78^\circ \text{C}$

Rise in bulk temperature of air, $\Delta T = T_{mo} - T_{mi}$

$$= 77.78 - 60$$

$$\Delta T = 17.78^\circ C$$

Heat transfer, $Q = mC_p (T_{mo} - T_{mi})$

$$= 0.015 \times 1005 (17.78^\circ C)$$

$$Q = 268.03 \text{ W}$$

Result:

1. $Q = 268.03 \text{ W}$
2. $\Delta T = T_{mo} - T_{mi} = 17.78^\circ C$

11. A vertical plate of 0.75 m height is at $170^\circ C$ and is exposed to air at a temperature of $105^\circ C$ and one atmosphere. Calculate:

1. **Mean heat transfer coefficient,**
2. **Rate of heat transfer per unit width of the plate.**

Given:

$$\text{Length, } L = 0.75 \text{ m}$$

$$\text{Wall temperature, } T_w = 170^\circ C$$

$$\text{Fluid temperature, } T_\infty = 105^\circ C$$

To find:

1. Heat transfer coefficient, (h)
2. Heat transfer (Q) per unit width.

Solution:

Velocity (U) is not given. So this is natural convection type problem.

$$\begin{aligned}\text{Film temperature, } T_f &= \frac{T_w + T_\infty}{2} \\ &= \frac{170 + 105}{2}\end{aligned}$$

$$T_f = 137.5^\circ\text{C}$$

Properties of air at $T_f = 137.5^\circ\text{C} \approx 140^\circ\text{C}$

[From HMT data Book, Page No. 33]

$$\text{Density, } \rho = 0.854 \text{ kg/m}^3$$

$$\text{Kinematic viscosity, } \nu = 27.80 \times 10^{-6} \text{ m}^2/\text{s}$$

$$\text{Prandtl Number } Pr = 0.684$$

$$\text{Thermal conductivity, } k = 0.03489 \text{ W/mK}$$

We know that,

$$\text{Coefficient of thermal expansion, } \beta = \frac{1}{T_f \text{ in } K}$$

$$\begin{aligned} \Rightarrow \beta &= \frac{1}{137.5 + 273} \\ &= \frac{1}{410.5} \end{aligned}$$

$$\Rightarrow \beta = 2.4 \times 10^{-3} \text{ K}^{-1}$$

We know that,

$$\text{Grashof Number, } Gr = \frac{g \times \beta \times L^3 \times \Delta T}{\nu^2}$$

[From HMT data Book, Page No. 134]

$$\Rightarrow Gr = \frac{9.81 \times 2.4 \times 10^{-3} \times (0.75)^3 \times (170 - 105)}{(27.80 \times 10^{-6})^2}$$

$$\Rightarrow \boxed{Gr = 8.35 \times 10^8}$$

$$\Rightarrow Gr Pr = 8.35 \times 10^8 \times 0.684$$

$$\boxed{Gr Pr = 5.71 \times 10^8}$$

Since $Gr Pr > 10^9$, flow is laminar.

$Gr Pr$ value is in between 10^4 and 10^9 i.e., $10^4 < Gr Pr < 10^9$

So, Nusselt Number

$$Nu = 0.59 (Gr Pr)^{0.25}$$

[From HMT data Book, Page No. 134]

$$\Rightarrow = 0.59 (5.71 \times 10^8)^{0.25}$$

$$\boxed{Nu = 91.21}$$

We know that,

$$\text{Nusselt Number, } Ne = \frac{hL}{k}$$

$$91.21 = \frac{h \times 0.75}{0.03489}$$

$$h = 4.24 \text{ W / m}^2 \text{ K}$$

$$\boxed{\text{Heat transfer coefficient, } h = 4.24 \text{ W/m}^2 \text{ K}}$$

We know,

$$\text{Heat transfer, } Q = hA(T_w - T_\infty)$$

$$= h \times W \times L \times (T_w - T_\infty)$$

$$= 4.24 \times 1 \times 0.75 \times (170 - 105)$$

$$[\because W = 1 \text{ m}]$$

$$\boxed{Q = 206.8 \text{ W}}$$

Result:

1. Heat transfer coefficient, $h = 4.24 \text{ W/m}^2\text{K}$
2. Heat transfer, $Q = 206.8 \text{ W}$

12. A vertical plate of 0.7 m wide and 1.2 m height maintained at a temperature of 90°C in a room at 30°C. Calculate the convective heat loss.

Given:

Wide, $W = 0.7 \text{ m}$

Height (or) Length, $L = 1.2 \text{ m}$

Wall temperature, $T_w = 90^\circ\text{C}$

Room temperature, $T_\infty = 30^\circ\text{C}$

To find:

Convective heat loss (Q)

Solution:

Velocity (U) is not given. So, this is natural convection type problem.

We know that,

$$\begin{aligned} \text{Film temperature, } T_f &= \frac{T_w + T_\infty}{2} \\ &= \frac{90 + 30}{2} \end{aligned}$$

$T_f = 60^\circ\text{C}$

Properties of air at 60°C:

[From HMT data Book, Page No. 33]

$$\rho = 1.060 \text{ kg/m}^3$$

$$\nu = 18.97 \times 10^{-6} \text{ m}^2/\text{s}$$

$$\text{Pr} = 0.696$$

$$K = 0.02896 \text{ W/mK}$$

We know,

$$\text{Coefficient of thermal expansion } \beta = \frac{1L}{T_f \text{ in K}}$$

$$\beta = \frac{1}{60 + 273} = 3 \times 10^{-3} \text{ K}^{-1}$$

$$\boxed{\beta = 3 \times 10^{-3} \text{ K}^{-1}}$$

$$\text{Grashor Number, Gr} = \frac{g \times \beta \times L^3 \times \Delta T}{\nu^2}$$

[From HMT data Book, Page No. 134]

$$= \frac{9.81 \times 3 \times 10^{-3} \times (1.2)^3 \times (90 - 30)}{(18.97 \times 10^{-6})^2}$$

$$\boxed{\text{Gr} = 8.4 \times 10^9}$$

$$\text{Gr Pr} = 8.4 \times 10^9 \times 0.696$$

$$\boxed{\text{Gr Pr} = 5.9 \times 10^9}$$

Since $\text{Gr Pr} > 10^9$, flow is turbulent.

For turbulent flow,

$$\text{NUSSELT Number, Nu} = 0.10 (\text{Gr Pr})^{0.333}$$

[From HMT data Book, Page No. 135]

$$\text{Nu} = 0.10 [5.9 \times 10^9]^{0.333}$$

$$\boxed{\text{Nu} = 179.3}$$

We know that,

$$\text{Local Nusselt Number, Nu} = \frac{hL}{k}$$

$$179.3 = \frac{h \times 1.2}{0.02896}$$

$$\text{Convective heat transfer coefficient } h = 4.32 \text{ W/m}^2\text{K}$$

$$\text{Heat loss, } Q = hA(\Delta T)$$

$$\begin{aligned} &= h \times W \times L \times (T_w - T_\infty) \\ &= 4.32 \times 0.7 \times 1.2 \times (90 - 30) \end{aligned}$$

$$Q = 218.16 \text{ W}$$

Result:

$$\text{Convective heat loss, } Q = 218.16 \text{ W}$$

13. A horizontal plate of 800 mm long, 70 mm wide is maintained at a temperature of 140°C in a large tank of full of water at 60°C. Determine the total heat loss from the plate.

Given:

$$\text{Horizontal plate length, } L = 800 \text{ mm} = 0.8 \text{ m}$$

$$\text{Wide, } W = 70 \text{ mm} = 0.070 \text{ m}$$

$$\text{Plate temperature, } T_w = 140^\circ\text{C}$$

$$\text{Fluid temperature, } T_\infty = 60^\circ\text{C}$$

To find:

Solution:

$$\begin{aligned} \text{Film temperature, } T_f &= \frac{T_w + T_\infty}{2} \\ &= \frac{140 + 60}{2} \end{aligned}$$

$$T_f = 100^\circ\text{C}$$

Properties of air at 100°C:

[From HMT data Book, Page No. 21]

$$\rho = 961 \text{ kg / m}^3$$

$$\nu = 0.293 \times 10^{-6} \text{ m}^2/\text{s}$$

$$\text{Pr} = 1.740$$

$$K = 0.6804 \text{ W/mK}$$

$$\beta_{(\text{water})} = 0.76 \times 10^{-3} \text{ K}^{-1}$$

[From HMT data Book, Page No. 21]

$$\text{Grahof Number, Gr} = \frac{g \times \beta \times L_c^3 \times \Delta T}{\nu^2} \dots\dots\dots (1)$$

For horizontal plate,

$$L_c = \text{Characteristic length} = \frac{W}{2}$$

$$L_c = \frac{0.070}{2}$$

$$= 0.035 \text{ m}$$

$$L_c = 0.035 \text{ m}$$

$$(1) \Rightarrow \text{Gr} = \frac{9.81 \times 0.76 \times 10^{-3} \times (0.035)^3 \times (140 - 60)}{(0.293 \times 10^{-6})^2}$$

$$\text{Gr} = 0.297 \times 10^9$$

$$\text{Gr Pr} = 0.297 \times 10^9 \times 1.740$$

$$\text{Gr Pr} = 0.518 \times 10^9$$

Gr Pr value is in between 8×10^6 and 10^{11} ,

i.e., $8 \times 10^6 < \text{Gr Pr} < 10^{11}$

so, for horizontal plate, upper surface heated,

$$\text{NUSSELT Number, } Nu = 0.15 (\text{Gr Pr})^{0.333}$$

[From HMT data Book, Page No. 135]

$$Nu = 0.15 [0.518 \times 10^9]^{0.333}$$

$$Nu = 119.66$$

We know that,

$$\text{Local Nusselt Number, } Nu = \frac{hL}{k}$$

$$119.66 = \frac{h_u \times 0.035}{0.6804}$$

$$\text{Heat transfer coefficient for upper surface heated, } h_u = 2326.19 \text{ W/m}^2\text{K}$$

For horizontal plate,

Lower surface heated,

[From HMT data Book, Page No. 136]

$$\text{Nusselt Number, } Nu = 0.27 [\text{Gr Pr}]^{0.25}$$

$$Nu = 0.27 [0.518 \times 10^9]^{0.25}$$

$$Nu = 40.73$$

We know that,

$$\text{Nusselt Number, } Nu = \frac{h_1 L_c}{k}$$

$$40.73 = \frac{h_l \times 0.035_c}{0.6804}$$

$$\text{Heat transfer coefficient for lower surface heated, } h_l = 791.79 \text{ W/m}^2\text{K}$$

$$\text{Total heat transfer, } Q = (h_u + h_l) A \Delta T$$

$$= (h_u + h_l) \times W \times L \times (T_w - T_\infty)$$

$$= 4.32 \times 0.7 \times 1.2 \times (90 - 30)$$

$$\Rightarrow Q = [2326.19 + 791.79] \times [0.070 \times 0.8] \times [140 - 60]$$

$Q = 13,968.55 \text{ W}$

Result:

Total heat loss, $Q = 13,968.55 \text{ W}$.

14. A steam pipe 80 mm in diameter is covered with 30 mm thick layer of insulation which has a surface emissivity of 0.94. The insulation surface temperature is 85°C and the pipe is placed in atmospheric air at 15°C . If the heat is lost both by radiation and free convection, find the following:

1. The heat loss from 5 m length of the pipe.
2. The overall heat transfer coefficient.
3. Heat transfer coefficient due to radiation.

Given:

Diameter of pipe = 80 mm

$$= 0.080 \text{ m}$$

Insulation thickness = 30 mm = 0.030 m

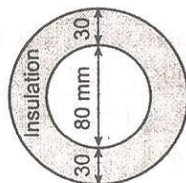
Actual diameter of the pipe, $D = 0.080 + 2 \times 0.030$

$$= 0.14 \text{ m}$$

Emissivity, $\varepsilon = 0.94$

Tube surface temperature, $T_w = 85^\circ\text{C}$

Air temperature, $T_\infty = 15^\circ\text{C}$



To find:

1. Heat loss from 5 m length of the pipe, Q
2. Overall heat transfer coefficient, h_t
3. Heat transfer coefficient due to radiation, h_r

Solution:

$$\text{Film temperature, } T_f = \frac{T_w + T_\infty}{2}$$

$$\frac{85 + 15}{2}$$

$$T_f = 50^\circ\text{C}$$

Properties of air at 50°C :

[From HMT data Book, Page No. 33]

$$\rho = 1.093 \text{ kg/m}^3$$

$$\nu = 17.95 \times 10^{-6} \text{ m}^2/\text{s}$$

$$\text{Pr} = 0.698$$

$$K = 0.02826 \text{ W/mK}$$

$$\text{Coefficient of thermal expansion } \beta = \frac{1}{T_f \text{ in K}}$$

$$\frac{1}{50 + 273}$$

$$\beta = 3.095 \times 10^{-3} \text{ K}^{-1}$$

We know that,

$$\text{Grashof number, Gr} = \frac{g \times \beta \times D^3 \times \Delta T}{\nu^2}$$

[From HMT data Book, Page No. 134]

$$\frac{9.81 \times 3.095 \times 10^{-3} \times (0.14)^3 \times (85 - 15)}{(17.95 \times 10^{-6})^2}$$

$$\boxed{\text{Gr} = 18.10 \times 10^6}$$

$$\text{Gr Pr} = 18.10 \times 10^6 \times 0.698$$

$$\boxed{\text{Gr Pr} = 1.263 \times 10^7}$$

For horizontal cylinder,

$$\text{Nusselt Number, Nu} = C [\text{Gr Pr}]^m$$

[From HMT data Book, Page No. 137]

$$\text{Gr Pr} = 1.263 \times 10^7$$

Corresponding $C = 0.125$, and $m = 0.333$

$$\text{Nu} = 0.125 [1.263 \times 10^7]^{0.333}$$

$$\boxed{\text{Nu} = 28.952}$$

We know that,

$$\text{Nu} = \frac{hD}{k}$$

$$28.952 = \frac{h \times 0.14}{0.02826}$$

$$h = 5.84 \text{ W/m}^2\text{K}$$

$$\boxed{\text{Convective heat transfer coefficient, } h_c = 5.84 \text{ W/m}^2\text{K}}$$

Heat lost by convection,

$$\begin{aligned}Q_{conv} &= hA(\Delta T) \\&= h \times \pi D \times L \times (T_w - T_\infty) \\&= 5.84 \times \pi 0.14 \times 5(85 - 15)\end{aligned}$$

$$Q_{conv} = 898.99 \text{ W}$$

Heat lost by radiation,

$$Q_{rad} = \varepsilon \sigma A [T_w^4 - T_\infty^4]$$

Where, ε = Emissivity

A = Area – m²

σ = Stefan Boltzmann constant

$$= 5.67 \times 10^{-8} \text{ W/m}^2 \text{ K}^4$$

T_w = Surface temperature, K

T_∞ = Fluid temperature, K

$$T_w = 85 + 273 \quad T_\infty = 15 + 273$$

$$T_w = 358 \text{ K}$$

$$T_\infty = 288$$

$$\Rightarrow Q_{rad} = \varepsilon \times \sigma \times \pi \times DL \times [T_w^4 - T_\infty^4]$$

$$= 0.94 \times 5.67 \times [10^{-8} \times \pi \times 0.14 \times 5 \times [358^4 - 288^4]]$$

$$Q_{rad} = 1118.90 \text{ W}$$

Total heat transfer, $Q_t = Q_{conv} + Q_{rad}$

$$= 898.99 + 111.90$$

$$Q_t = 2017.89 \text{ W}$$

Total heat transfer, $Q_t = h_t A \Delta T$

$$= h_t \times \pi DL \times (T_w - T_\infty)$$

$$2017.89 = h_t \times \pi \times 0.14 \times 5 \times (85 - 15)$$

$$\Rightarrow h_t = 13.108 \text{ W/m}^2\text{K}$$

$$\text{Overall heat transfer coefficient, } h_t = 13.108 \text{ W/m}^2\text{K}$$

Radiative heat transfer coefficient,

$$h_r = h_t - h_c$$

$$= 13.108 - 5.84$$

$$h_r = 7.268 \text{ W/m}^2\text{K}$$

Result:

1. Heat loss from 5 m length of pipe
 - (i) By convection, $Q_c = 898.99 \text{ W}$
 - (ii) By radiation, $Q_r = 1118.90 \text{ W}$
2. Overall heat transfer coefficient, $h_t = 13.108 \text{ W/m}^2\text{K}$
3. Radiative heat transfer coefficient, $h_r = 7.268 \text{ W/m}^2\text{K}$

15. A vertical plate of 40 cm long is maintained at 80°C and is exposed to air at 22°C . Calculate the following:

1. **Boundary layer thickness at the trailing edge of the plate.**
2. **The same plate is placed in a wind tunnel and air is blown over it at a velocity of 5 m/s. Calculate boundary layer thickness.**
3. **Average heat transfer coefficient for natural and forced convection for the above mentioned data.**

Given:

Length, $L = 40 \text{ cm} = 0.40 \text{ m}$

Plate temperature, $T_w = 80^\circ\text{C}$

Fluid temperature, $T_\infty = 22^\circ\text{C}$

To find: Case (i)

- (i) Boundary layer thickness (Natural convection).

Case (ii)

- (i) Boundary layer thickness at velocity $U = 5 \text{ m/s}$ (Forced convection).
(ii) Average heat transfer coefficient for forced convection,

Solution:

We know that,

$$\begin{aligned} \text{Film temperature, } T_f &= \frac{T_w + T_\infty}{2} \\ &= \frac{80 + 22}{2} \end{aligned}$$

$T_f = 51^\circ\text{C}$

Properties of air at 51°C : $\approx 50^\circ\text{C}$:

[From HMT data Book, Page No. 33]

$$\rho = 1.093 \text{ kg/m}^3$$

$$\nu = 17.95 \times 10^{-6} \text{ m}^2/\text{s}$$

$$\text{Pr} = 0.698$$

$$K = 0.02826 \text{ W/mK}$$

$$\beta = \frac{1}{T_f \text{ in K}}$$

$$= \frac{1}{51 + 273}$$

$$\beta = 3.086 \times 10^{-3}$$

Case (i): for free convection,

$$\text{Gr} = \frac{g \times \beta \times L^3 \times \Delta T}{\nu^2}$$

[From HMT data Book, Page No. 134]

$$= \frac{9.81 \times 3.086 \times 10^{-3} \times (0.14)^3 \times (80 - 22)}{(17.95 \times 10^{-6})^2}$$

$$\text{Gr} = 3.48 \times 10^8$$

$$\text{Gr Pr} = 3.48 \times 10^8 \times 0.698$$

$$\text{Gr Pr} = 2.43 \times 10^6$$

Since $\text{Gr} < 10^9$, flow is laminar,

For free convection, laminar flow:

$$\text{Boundary layer thickness, } \delta_x = \left[3.93 \times (\text{Pr})^{-0.5} \times (0.952 + \text{Pr})^{0.25} \times (\text{Gr})^{-0.25} \right] \times x =$$

[From HMT data Book, Page No. 134]

$$\Rightarrow \delta_x = \left[3.93 \times (0.698)^{-0.5} \times (0.952 + 0.698)^{0.25} \times (3.48 \times 10^8)^{-0.25} \right] \times 0.4$$

$$[\because x = L = 0.40\text{m}]$$

$$\Rightarrow \delta_x = 0.0156\text{m}$$

Case (ii): For forced convection,

$$\begin{aligned}\text{Reynolds number, Re} &= \frac{UL}{\nu} \\ &= \frac{5 \times 0.40}{17.95 \times 10^{-6}} \\ \boxed{\text{Re} = 1.11 \times 10^5} &\quad \dots\dots (1)\end{aligned}$$

Since $\text{Re} < 5 \times 10^5$ flow is laminar.

For forced convection, laminar flow:

$$\begin{aligned}\text{Boundary layer thickness } \delta_x \text{ or } \delta_{hx} &= 5 \times x \times (\text{Re})^{-5/4} \\ &\quad \text{[From HMT data Book, Page No. 134]}\end{aligned}$$

$$\begin{aligned}\delta_x &= 5 \times 0.40 \times (1.11 \times 10^5)^{-5/4} \\ &\quad [\because x = L = 0.40\text{m}] \\ \delta_x &= 6.003 \times 10^{-3} \text{ m} \quad \dots\dots (2)\end{aligned}$$

From equation (1) and (2), we know that, boundary layer thickness in forced convection is less than that in free convection.

Case (iii): Average heat transfer coefficient for natural convection, h :

For free convection, laminar flow, vertical plate:

[From HMT data Book, Page No. 135]

$$\begin{aligned}\text{Nusselt number, Nu} &= 0.59(\text{Gr Pr})^{0.25} \\ &= 0.59 (2.43 \times 10^6)^{0.25} \\ \boxed{\text{Nu} = 23.29}\end{aligned}$$

We know that,

$$\begin{aligned}\text{Nu} &= \frac{hL}{k} \\ 23.29 &= \frac{h \times 0.4}{0.02826}\end{aligned}$$

$$h = 1.645 \text{ W/m}^2\text{K}$$

..... (3)

Average heat transfer coefficient for forced convection, h :

For forced convection, laminar flow, flat plate:

Local Nusselt number,

$$\text{Nu}_x = \frac{h_x L}{k}$$

$$98.13 = \frac{h_x 0.4}{0.02826}$$

Local heat transfer coefficient $h_x = 6.932 \text{ W/m}^2 \text{ K}$

Average heat transfer coefficient $h = 2 \times 6.932$

$$h = 13.86 \text{ W/m}^2\text{K}$$

..... (4)

From equation (3) and (4) we know that heat transfer coefficient in forced convection is much larger than that in free convection.

Result:

Case (i): 1. $\delta_x (\text{Natural convection}) = 0.0156 \text{ m}$

Case (ii): 1. $\delta_x (\text{Forced convection}) = 6.003 \times 10^{-3} \text{ m}$

Case (iii): 1. $h (\text{Natural convection}) = 1.645 \text{ W/m}^2\text{K}$

2. $h (\text{Forced convection}) = 13.86 \text{ W/m}^2\text{K}$

16. Water is to be boiled at atmospheric pressure in a polished copper pan by means of an electric heater. The diameter of the pan is 0.38 m and is kept at 115°C. Calculate the following

- 1. Power required to boil the water**
- 2. Rate of evaporation**
- 3. Critical heat flux.**

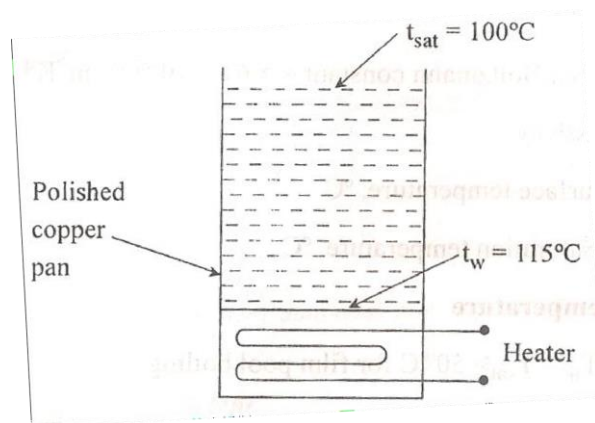
Given:

Diameter, $d = 0.38 \text{ m}$;

Surface temperature, $T_w = 115^\circ\text{C}$

To find:

1. Power required, (P)
2. Rate of evaporation, $(\dot{m})$
3. Critical heat flux, $(\frac{Q}{A})$



Solution:

We know that, saturation temperature of water is 100°C .

i.e.

$T_{\text{sat}} = 100^\circ\text{C}$

Density, $\rho_l = 961 \text{ kg/m}^3$

Kinematic viscosity, $\nu = 0.293 \times 10^{-6} \text{ m}^2/\text{s}$

Prandtl Number, $Pr = 1.740$

Specific heat, $C_{pl} = 4216 \text{ J/kg K}$

Dynamic viscosity, $\mu_l \times \nu = 961 \times 0.293 \times 10^{-6}$
 $= 281.57 \times 10^{-6} \text{ Ns/m}^2$

From Steam Table

At 100°C

Enthalpy of evaporation, $h_{fg} = 2256.9 \text{ kJ/kg}$

$$h_{fg} = 2256.9 \times 10^3 \text{ J/kg}$$

Specific volume of vapour, $\nu_g = 1.673 \text{ m}^3/\text{kg}$

Density of vapour, $\rho_v = \frac{1}{\nu_g}$

$$= \frac{1}{1.673}$$

$$\rho_v = 0.597 \text{ kg/m}^3$$

$$\Delta T = \text{Excess temperature} = T_w - T_{sat} = 115^\circ\text{C} - 100 = 15^\circ\text{C}$$

$$\Delta T = 15^\circ\text{C} < 50^\circ\text{C}. \text{ So, this is nucleate pool boiling process.}$$

1. Power required to boil the water

For Nucleate pool boiling

$$\text{Heat flux, } \frac{Q}{A} = \mu_l \times h_{fg} \left[\frac{g \times (p - p_v)}{\sigma} \right]^{0.5} \times \left[\frac{C_{pl} \times \Delta T}{C_{sf} \times h_{fg} Pr^n} \right]^3 \quad \dots\dots (1)$$

Where σ = surface tension for liquid vapour interface

At 100°C

$$\sigma = 0.0588 \text{ N/m}$$

For water – copper ==> C_{sf} = surface fluid constant = 0.013

$n = 1$ for water

Substitute

$\mu_l, h_{fg}, p_l, p_v, \sigma, C_{pl}, \Delta T, C_{sf}, h_{fg}, n$ And P_r values in Equn (1)

$$(1) \Rightarrow \frac{Q}{A} = 281.57 \times 10^{-6} \times 2256.9 \times 10^3 \times \left[\frac{9.81 \times (961 - 0.597)}{0.0588} \right]^{0.5} \times \left[\frac{4216 \times 15}{0.013 \times 2256.9 \times 10^3 \times (1.74)^+} \right]^3$$

$$\text{Heat flux, } \frac{Q}{A} = 4.83 \times 10^5 \text{ W / m}^2$$

$$\Rightarrow \text{Heat transfer, } Q = 4.83 \times 10^5 \times A$$

$$= 4.83 \times 10^5 \times \frac{\pi}{4} d^2$$

$$= 4.83 \times 10^5 \times \frac{\pi}{4} (0.38)^2$$

$$Q = 54.7 \times 10^3 \text{ W}$$

$$Q = 54.7 \times 10^3 = P$$

$$\Rightarrow \text{Power} = 54.7 \times 10^3 \text{ W}$$

2. Rate of evaporation, (m)

We know that,

$$\text{Heat transferred, } Q = m \times h_{fg}$$

$$\Rightarrow m = \frac{Q}{h_{fg}}$$

$$= \frac{54.7 \times 10^3}{2256.9 \times 10^3}$$

$$m = 0.024 \text{ kg/s}$$

3. Critical heat flux

For Nucleate pool boiling, critical heat flux,

$$\begin{aligned} \frac{Q}{A} &= 0.18 h_{fg} \times p_v \left[\frac{\sigma \times g \times (\rho_l - \rho_v)}{p_v^2} \right]^{0.25} \\ &= 0.18 \times 2256.9 \times 10^3 \times 0.597 \times \left[\frac{0.0588 \times 9.81 \times (961 - 0.597)}{(0.597)^2} \right]^{0.25} \\ \frac{Q}{A} &= 1.52 \times 10^6 \text{ W / m}^2 \end{aligned}$$

$$\text{Critical Heat flux, } q = \frac{Q}{A} = 1.52 \times 10^6 \text{ W / m}^2$$

Result:

1. $P = 54.7 \times 10^3 \text{ W}$
2. $M = 0.024 \text{ kg/s}$
3. $\frac{Q}{A} = q = 1.52 \times 10^6 \text{ W / m}^2$.

17. Water is boiling on a horizontal tube whose wall temperature is maintained at 15°C above the saturation temperature of water. Calculate the nucleate boiling heat transfer co-efficient. Assume the water to be at a pressure of 10 atm. And also find the change in value of heat transfer co-efficient when

1. *The temperature difference is increased to 30°C at a pressure of 10 atm.*
2. *The pressure is raised to 20 atm at $\Delta T = 15^\circ \text{C}$.*

Given:

Wall temperature is maintained at 15°C above the saturation temperature.

$$T_w = 115^\circ \text{C} \quad \left[\because T_{\text{sat}} = 100^\circ \text{C}; T_w = 100 + 15 = 115^\circ \text{C} \right]$$

$$P = 10 \text{ atm} = 10 \text{ bar}$$

Case (i):

$$\Delta T = 30^\circ \text{C}; p = 10 \text{ atm} = 10 \text{ bar}$$

Case (ii):

$$P = 20 \text{ atm} = 20 \text{ bar}; \Delta T = 15^\circ \text{C}$$

Solution:

We know that, for horizontal surface, heat transfer co-efficient

$$\begin{aligned} h &= 5.56(\Delta T)^3 \\ \Rightarrow h &= 5.56(T_w - T_{\text{sat}})^3 \\ &= 5.56(115 - 100)^3 \end{aligned}$$

$h = 18765 \text{ W/m}^2\text{K}$

Heat transfer co-efficient other than atmospheric pressure

$$\begin{aligned} h_p &= h p^{0.4} \\ &= 18765 \times [10]^{0.4} \end{aligned}$$

Heat transfer co-efficient, $h_p = 47.13 \times 10^3 \text{ W / m}^2 \text{ K}$

Case (i)

$$p = 10 \text{ bar}; \Delta T = 30^\circ \text{C}$$

$$h = 150 \times 10^3 \text{ W/m}^2\text{K}$$

Heat transfer co-efficient other than atmospheric pressure

$$h_p = h_p^{0.4}$$
$$= 150 \times 10^3 (10)^{0.4}$$

$$h_p = 377 \times 10^3 \text{ W/m}^2 \text{ K}$$

Case (ii)

$$p = 20 \text{ bar}; \Delta T = 15^\circ \text{ C}$$

$$\text{Heat transfer co-efficient, } h = 5.56 (\Delta T)^3 = 5.56 (15)^3$$

$$h = 18765 \text{ W/m}^2 \text{ K}$$

Heat transfer co-efficient other than atmospheric pressure

$$h_p = h_p^{0.4}$$
$$= 150 \times (10)^{0.4}$$

$$h_p = 62.19 \times 10^3 \text{ W/m}^2 \text{ K}$$

Result:

Nucleate boiling heat transfer co-efficient

$$h_p = 47.13 \times 10^3 \text{ W/m}^2 \text{ K}$$

Case (i)

$$h_p = 377 \times 10^3 \text{ W/m}^2 \text{ K}$$

Case (ii)

$$h_p = 62.19 \times 10^3 \text{ W/m}^2 \text{ K}$$

18. A vertical flat plate in the form of fin is 500 mm in height and is exposed to steam at atmospheric pressure. If surface of the plate is maintained at 60°C, calculate the following

- **The film thickness at the trailing edge**
- **Overall heat transfer co-efficient**
- **Heat transfer rate**
- **The condensate mass flow rate.**

Given:

Height (or) Length, $L = 500 \text{ mm } 0.5 \text{ m}$

Surface temperature, $T_w = 60^\circ\text{C}$

To find:

1. δ_x
2. h
3. Q
4. m

Solution:

We know that, saturation temperature of water is 100°C

i.e., $T_{\text{sat}} = 100^\circ\text{C}$

$$h_{\text{fg}} = 2256.9 \text{ kJ/kg}$$

$$h_{\text{fg}} = 2256.9 \times 10^3 \text{ J/kg}$$

We know that,

$$\begin{aligned} \text{Film temperature, } T_f &= \frac{T_w + T_\infty}{2} \\ &= \frac{60 + 100}{2} \\ &= 80^\circ\text{C} \end{aligned}$$

$T_f = 80^\circ\text{C}$

Properties of air at 80°C

$$\rho = 974 \text{ kg / m}^3$$

$$\nu = 0.364 \times 10^{-6} \text{ m}^2/\text{s}$$

$$k = 0.6687 \text{ W/mK}$$

$$\mu = \rho \times \nu = 974 \times 0.364 \times 10^{-6}$$

$$\mu = 354.53 \times 10^{-6} \text{ Ns / m}^2$$

1 Film thickness (δ_x)

We know for vertical plate,

$$\text{Film thickness, } \delta_x = \left[\frac{4 \mu k x (T_{\text{sat}} - T_w)}{g \times h_{fg} \times \rho^2} \right]^{0.25}$$

Where

$$x = L = 0.5 \text{ m}$$

$$\Rightarrow \delta_x = \frac{4 \times 354.53 \times 10^{-6} \times 0.6687 \times 0.5 \times (100 - 60)}{9.81 \times 2256.9 \times 10^3 \times (974)^2}$$

$$\delta_x = 1.73 \times 10^{-4} \text{ m}$$

2 Average heat transfer co-efficient, (h)

For vertical surface, Laminar flow

$$h = 0.943 \left[\frac{k^3 \times \rho^2 \times g \times h_{fg}}{\mu \times L \times (T_{\text{sat}} - T_w)} \right]^{0.25}$$

The factor 0.943 may be replaced by 1.13 for more accurate result as suggested by Mc Adams

Since $Gr < 10^9$, flow is laminar,

$$h = 1.13 \left[\frac{(0.6687)^3 \times (974)^2 \times 981 \times 2256.9 \times 10^3}{354.53 \times 10^{-6} \times 0.5 \times (100 - 60)} \right]^{0.25}$$

$$h = 6164.3 \text{ W/m}^2/\text{K}$$

3 Heat transfer rate, (Q)

$$\text{Heat transfer, } Q = hA(T_{sat} - T_w)$$

$$= h \times L \times W \times (T_{sat} - T_w)$$

$$= 6164.3 \times 0.5 \times 1 \times (100 - 60)$$

$$Q = 1, 23,286 \text{ W}$$

4 **Condensate mass flow rate, (m)**

We know that,

$$Q = m \times h_{fg}$$

$$\Rightarrow m = \frac{Q}{h_{fg}}$$

$$m = \frac{1,23,286}{2256.9 \times 10^3}$$

$$m = 0.054 \text{ kg/s}$$

Result:

1. $\delta_x = 1.73 \times 10^{-4}$
2. $h = 6164.3 \text{ W/m}^2/\text{K}$
3. $Q = 123286 \text{ W}$
4. $m = 0.054 \text{ kg/s}$

19. Steam at 0.080 bar is arranged to condense over a 50 cm square vertical plate. The surface temperature is maintained at 20°C. Calculate the following.

- a. **Film thickness at a distance of 25 cm from the top of the plate.**
- b. **Local heat transfer co-efficient at a distance of 25 cm from the top of the plate.**
- c. **Average heat transfer co-efficient**
- d. **Total heat transfer**
- e. **Total steam condensation rate**

f. What would be the heat transfer co-efficient if the plate is inclined at 30°C with horizontal plane.

Given:

Pressure, $p = 0.080 \text{ bar}$

Area, $A = 50 \text{ cm} \times 0.50 = 0.25 \text{ m}^2$

Surface temperature, $T_w = 20^\circ\text{C}$

Distance, $x = 25 \text{ cm}$

To find:

- a) δ_x
- b) h_x
- c) h
- d) Q
- e) m
- f) h at 30°C

Solution:

Properties of steam at 0.080 bar

$$T_{\text{sat}} = 41.53^\circ\text{C}$$

$$h_{\text{fg}} = 2403.2 \text{ kJ/kg} = 2403.2 \times 10^3 \text{ J/kg}$$

We know that,

$$\begin{aligned} \text{Film temperature, } T_f &= \frac{T_w + T_\infty}{2} \\ &= \frac{20 + 41.53}{2} \\ &= 30.76^\circ\text{C} \end{aligned}$$

Properties of air at 30.76°C : $\approx 30^\circ\text{C}$:

$$\rho = 997 \text{ kg / m}^3$$

$$\nu = 0.83 \times 10^{-6} \text{ m}^2/\text{s}$$

$$k = 0.612 \text{ W/mK}$$

$$\mu = p \times v = 997 \times 0.83 \times 10^{-6}$$

$$\mu = 827.57 \times 10^{-6} \text{ Ns / m}^2$$

a) Film thickness

We know, for vertical surfaces

$$\delta_x = \left[\frac{4 \mu k x (T_{sat} - T_w)}{g \times h_{fg} \times p^2} \right]^{0.25}$$

$$= \left[\frac{4 \times 827.57 \times 10^{-6} \times 0.612 \times 0.25 \times (41.53 - 20)}{9.81 \times 2403.2 \times 10^3 \times (997)^2} \right]$$

$$\delta_x = 1.46 \times 10^{-4} \text{ m}$$

b) Local heat transfer co-efficient (h_x)

$$h_x = \frac{k}{\delta_x}$$

$$h_x = \frac{0.612}{1.46 \times 10^{-4}}$$

$$h_x = 4.191 \text{ W/m}^2 \text{ K}$$

c) Average heat transfer co-efficient (h)

$$h = 0.943 \left[\frac{k^3 \times p^2 \times g \times h_{fg}}{\mu \times L (T_{sat} - T_w)} \right]^{0.25}$$

The factor 0.943 may be replaced by 1.13 for more accurate result as suggested by Mc Adams

$$\Rightarrow h = 1.13 \left[\frac{k^3 p^2 g h_{fg}}{\mu L (T_{sat} - T_w)} \right]^{0.25}$$

Where $L = 50 \text{ cm} = 0.5 \text{ m}$

$$h = 1.13 \left[\frac{(0.612)^3 \times (997)^2 \times 9.81 \times 2403.2 \times 10^3}{827.51 \times 10^{-6} \times 5(41.53 - 20)} \right]^{0.25}$$

$$h = 5599.6 \text{ W/m}^2\text{K}$$

d) Heat transfer (Q)

We know that,

$$\begin{aligned} Q &= hA(T_{sat} - T_w) \\ &= h \times A \times (T_{sat} - T_w) \\ &= (5599.6) \times 0.25 \times (41.53 - 20) \end{aligned}$$

$$Q = 30,139.8 \text{ W}$$

e) Total steam condensation rate (m)

Heat transfer, $Q = m \times h_{fg}$

$$\begin{aligned} \Rightarrow m &= \frac{Q}{h_{fg}} \\ m &= \frac{30,139.8}{2403.2 \times 10^3} \\ m &= 0.0125 \text{ kg/s} \end{aligned}$$

f) If the plate is inclined at

$$\begin{aligned} h_{\text{inclined}} &= h_{\text{vertical}} \times (\sin \theta)^{1/4} \\ \Rightarrow h_{\text{inclined}} &= h_{\text{vertical}} \times (\sin 30)^{1/4} \\ \Rightarrow h_{\text{inclined}} &= 5599.6 \times (1/2)^{1/4} \end{aligned}$$

$$h_{\text{inclined}} = 4,708.6 \text{ W/m}^2\text{K}$$

Let us check the assumption of laminar film condensation

We know that,

$$\text{Reynolds Number, } Re = \frac{4m}{W\mu}$$

Where

$$W = \text{width of the plate} = 50 \text{ cm} = 0.50 \text{ m}$$

$$\Rightarrow Re = \frac{4 \times 0.0125}{0.50 \times 827.51 \times 10^{-6}}$$

$$\boxed{Re = 120.8} < 1800$$

So our assumption (laminar flow) is correct.

Result:

- a) $\delta_x = 1.46 \times 10^{-4}$
- b) $h_x = 4191 \text{ W/m}^2\text{K}$
- c) $h = 5599.6 \text{ W/m}^2\text{K}$
- d) $Q = 30,139.8 \text{ W}$
- e) $m = .0125 \text{ kg/s}$
- f) $h_{\text{inclined}} = 4708.6 \text{ W/m}^2\text{K}$

20. Saturated steam at $t_{\text{sat}} = 100^\circ\text{C}$ condenses on the outer surface of a 1.4 m long, 2 m outer diameter vertical tube maintained at a uniform temperature $T_w = 60^\circ\text{C}$. Assuming film condensation, find the following.

- a) **Local heat transfer co-efficient at the bottom of the tube.**
- b) **Average heat transfer co-efficient over the entire length of the tube.**

Given:

Saturation temperature, $T_{\text{sat}} = 100^\circ\text{C}$

Length, $L = 1.4 \text{ m}$

Outer diameter, $D = 2 \text{ m}$

Surface temperature, $T_w = 60^\circ\text{C}$

To find:

1. Local heat transfer co-efficient h_x
2. Average heat transfer co-efficient, h

Solution:

Properties of steam at 100°C

Enthalpy of evaporation,

$$\begin{aligned}h_{fg} &= 2256.9 \text{ kJ} / \text{kg} \\&= 2256.9 \times 10^3 \text{ J} / \text{kg}\end{aligned}$$

We know that,

$$\begin{aligned}\text{Film temperature, } T_f &= \frac{T_w + T_\infty}{2} \\&= \frac{60 + 100}{2}\end{aligned}$$

$T_f = 80^\circ\text{C}$

Properties of air at 80°C :

$$\rho = 974 \text{ kg} / \text{m}^3$$

$$\nu = 0.364 \times 10^{-6} \text{ m}^2/\text{s}$$

$$k = 0.6687 \text{ W/mK}$$

$$\mu = \rho \times \nu = 974 \times 0.364 \times 10^{-6}$$

$$\mu = 354 \times 10^{-6} \text{ Ns} / \text{m}^2$$

Assuming that the condensate film is laminar

For vertical surfaces, laminar flow,

$$\text{Film thickness} = \delta_x = \left[\frac{4\mu kx(T_{\text{sat}} - T_w)}{g \times h_{fg} \times \rho^2} \right]^{0.25}$$

$$= \left[\frac{4 \times 354.53 \times 10^{-6} \times 0.6687 \times 1.4 (100 - 60)}{9.81 \times 2256.9 \times 10^3 \times (974)^2} \right]^{0.25}$$

$$[\because x = L = 1.4m]$$

Local heat transfer co-efficient (h_x)

$$h_x = \frac{k}{\delta_x}$$

$\delta_x = 1.46 \times 10^{-4} \text{ m}$

$$h_x = \frac{0.6687}{2.24 \times 10^{-4}}$$

$h_x = 2985.26 \text{ W/m}^2 \text{ K}$

Average heat transfer co-efficient (h)

$$h = 0.943 \left[\frac{k^3 \times p^2 \times g \times h_{fg}}{\mu \times L \times (T_{sat} - T_w)} \right]^{0.25}$$

The factor 0.943 may be replaced by 1.13 for more accurate result as suggested by Mc Adams

$$\Rightarrow h = 1.13 \left[\frac{k^3 p^2 g h_{fg}}{\mu L (T_{sat} - T_w)} \right]^{0.25}$$

$$h = 1.13 \left[\frac{(0.6687)^3 \times (974)^2 \times 9.81 \times 2256.9 \times 10^3}{354.53 \times 10^{-6} \times 1.4 (100 - 60)} \right]^{0.25}$$

$h = 4765.58 \text{ W/m}^2 \text{ K}$

Let us check the assumption of laminar film condensation

We know that,

$$\text{Reynolds Number, } Re = \frac{4m}{P\mu} \quad \dots\dots (1)$$

Heat transfer $Q = hA\Delta T$

$$\begin{aligned} &= h \times \pi DL \times (T_{sat} - T_w) \\ &= 4765.58 \times \pi \times 2 \times 1.4 \times (100 - 60) \end{aligned}$$

$$Q = 1.67 \times 10^6$$

We know that,

$$\begin{aligned} Q &= m_{hfg} \\ 1.67 \times 10^6 &= m(2256.9 \times 10^3) \end{aligned}$$

$$m = 0.739 \text{ kg/s} \quad \dots\dots (2)$$

Perimeter, $P = \pi D$

$$= \pi \times 2$$

$$P = 6.283 \text{ m} \quad \dots\dots (3)$$

Substitute P, m, μ values in equation (1)

$$(1) \Rightarrow Re = \frac{4 \times 0.739}{6.283 \times 354.53 \times 10^{-6}}$$

$$Re = 1327.04$$

So our assumption (laminar flow) is correct

Result:

1. Local heat transfer co-efficient, $h_x = 2985.26 \text{ W/m}^2\text{K}$
2. Average heat transfer co-efficient, $h = 4765.58 \text{ W/m}^2\text{K}$

TWO MARK QUESTIONS AND ANSWERS

1. What is dimensional analysis?

Dimensional analysis is a mathematical method which makes use of the study of the dimensions for solving several engineering problems. This method can be applied to all types of fluid resistances, heat flow problems in fluid mechanics and thermodynamics.

2. State Buckingham π theorem.

Buckingham π theorem states as follows: “If there are n variables in a dimensionally homogeneous equation and if these contain m fundamental dimensions, then the variables are arranged into $(n - m)$ dimensionless terms. These dimensionless terms are called π terms.

3. What are all the advantages of dimensional analysis?

- It expresses the functional relationship between the variables in dimensional terms.
- It enables getting up a theoretical solution in a simplified dimensionless form.
- The results of one series of tests can be applied to a large number of other similar problems with the help of dimensional analysis.

4. What are all the limitations of dimensional analysis?

- The complete information is not provided by dimensional analysis. It only indicates that there is some relationship between the parameters.
- No information is given about the internal mechanism of physical phenomenon.
- Dimensional analysis does not give any clue regarding the selection of variables.

5. Define Reynolds number (Re).

It is defined as the ratio of inertia force to viscous force.

$$Re = \frac{\text{Inertiaforce}}{\text{Viscousforce}}$$

6. Define Prandtl number (Pr)

It is the ratio of the momentum diffusivity to the thermal diffusivity.

$$Pr = \frac{\text{Momentumdiffusivity}}{\text{Thermaldiffusivity}}$$

7. Define Nusselt Number (Nu).

It is defined as the ratio of the heat flow by convection process under an unit temperature gradient to the heat flow rate by conduction under an unit temperature gradient through a stationary thickness (L) of metre.

$$\text{Nusselt Number (Nu)} = \frac{q_{conv}}{q_{cond}}$$

8. Define Grashof number (Gr)

It is defined as the ratio of product of inertia force and buoyancy force to the square of viscous force.

$$Gr = \frac{\text{Inertiaforce} \times \text{Buoyancyforce}}{(\text{Viscousforce})^2}$$

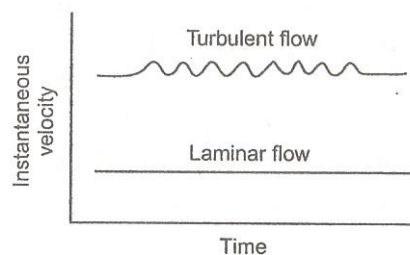
9. Define Stanton number (St).

It is the ratio of Nusselt number to the product of Reynolds number and Prandtl number.

$$St = \frac{Nu}{Re \times Pr}$$

10. What is meant by laminar flow and turbulent flow?

Laminar flow: Laminar flow is sometimes called stream line flow. In this type of flow, the fluid moves in layers and each fluid particle follows a smooth continuous path. The fluid particles in each layer remain in an orderly sequence without mixing with each other.



Turbulent flow: In addition to the laminar type of flow, a distinct irregular flow is frequently observed in nature. This type of flow is called turbulent flow. The path of any individual particle is zig-zag and irregular. Fig. shows the instantaneous velocity in laminar and turbulent flow.

11. Define convection.

Convection is a process of heat transfer that will occur between a solid surface and a fluid medium when they are at different temperatures.

12. State Newton's law of convection.

Heat transfer from the moving fluid to solid surface is given by the equation

$$Q = hA(T_w - T_\infty)$$

This equation is referred to as Newton's law of cooling.

Where

h - Local heat transfer coefficient in $\text{W/m}^2\text{K}$

A - Surface area in m^2

T_w - Surface (or) Wall temperature in K

T_∞ - Temperature of fluid in K

13. What is meant by free or natural convection?

If the fluid motion is produced due to change in density resulting from temperature gradients, the mode of heat transfer is said to be free or natural convection.

14. What is forced convection?

If the fluid motion is artificially created by means of an external forced like a blower or fan, that type of heat transfer is known as forced convection.

15. What is the form of equation used to calculate heat transfer for flow through cylindrical pipes?

$$\text{Nu} = 0.023 (\text{Re})^{0.8} (\text{Pr})^n$$

$n = 0.4$ for heating of fluids.

$n = 0.3$ for cooling of fluids.

16. What are the dimensionless parameters used in forced convection?

- Reynolds number (Re).
- Nusselt number (Nu).
- Prandtl number (Pr).

17. Define boundary layer thickness.

The thickness of the boundary layer has been defined as the distance from the surface at which the local velocity or temperature reaches 99% of the external velocity or temperature.

18. Indicate the concept or significance of boundary layer.

In the boundary layer concept the flow field over a body is divided into two regions;

- A thin region near the body called the boundary layer where the velocity and the temperature gradients are large.
- The region outside the boundary layer where the velocity and the temperature gradients are very nearly equal to their free stream values.

19. Write down the momentum equation for a steady, two dimensional flow of an incompressible, constant property Newtonian fluid in the rectangular coordinate system and mention the physical significance of each term.

Momentum equation,

$$\rho \left[u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} \right] = F_x - \frac{\partial P}{\partial x} + \mu \left[\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right]$$

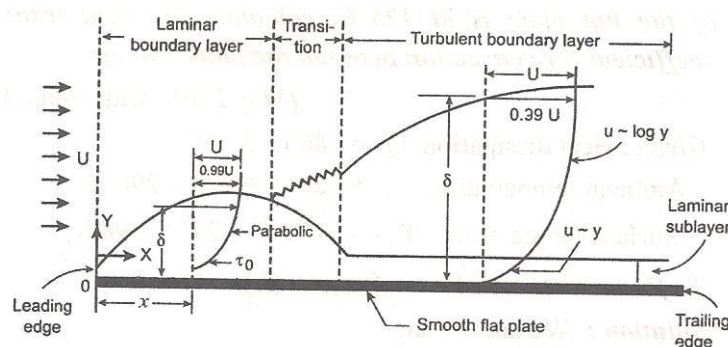
Where, $\rho \left[u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} \right] = \text{Inertia forces.}$

$F_x = \text{Body force.}$

$$\frac{\partial P}{\partial x} = \text{Pressure force.}$$

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = \text{Viscous forces.}$$

20. Sketch the boundary development of a flow.



21. Define displacement thickness.

The displacement thickness is the distance, measured perpendicular to the boundary, by which the free stream is displaced on account of formation of boundary layer.

22. Define momentum thickness.

The momentum thickness is defined as the distance through which the total loss of momentum per second be equal to if it were passing a stationary plate.

23. Define energy thickness.

The energy thickness can be defined as the distance, measured perpendicular to the boundary of the solid body, by which the boundary should be displaced to compensate for the reduction in kinetic energy of the flowing fluid on account of boundary layer formation.